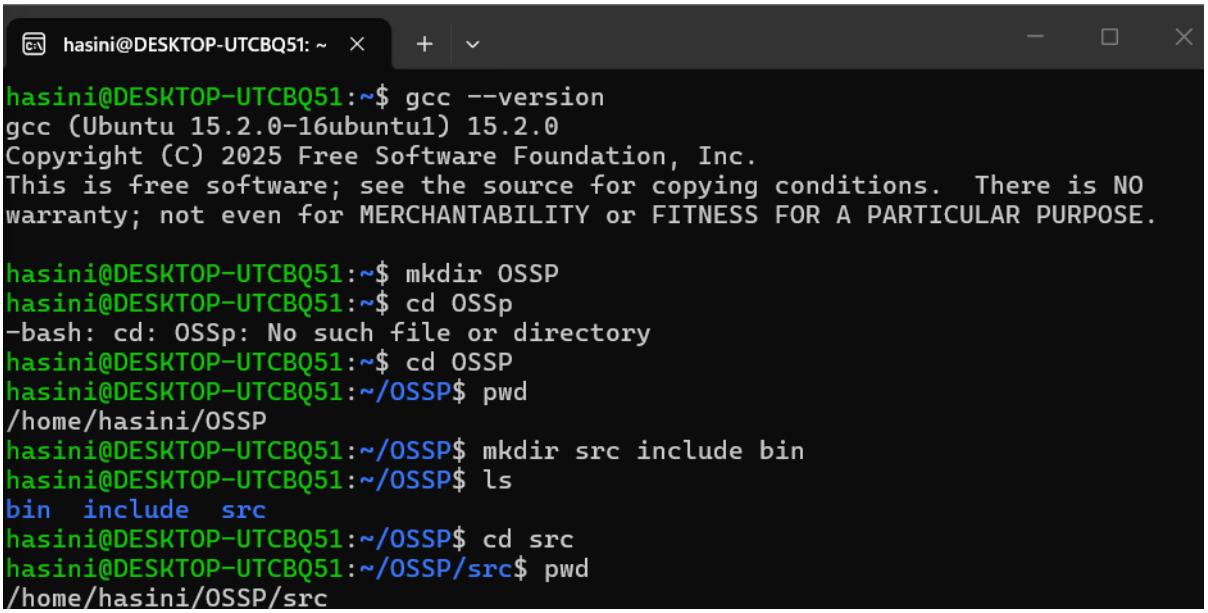
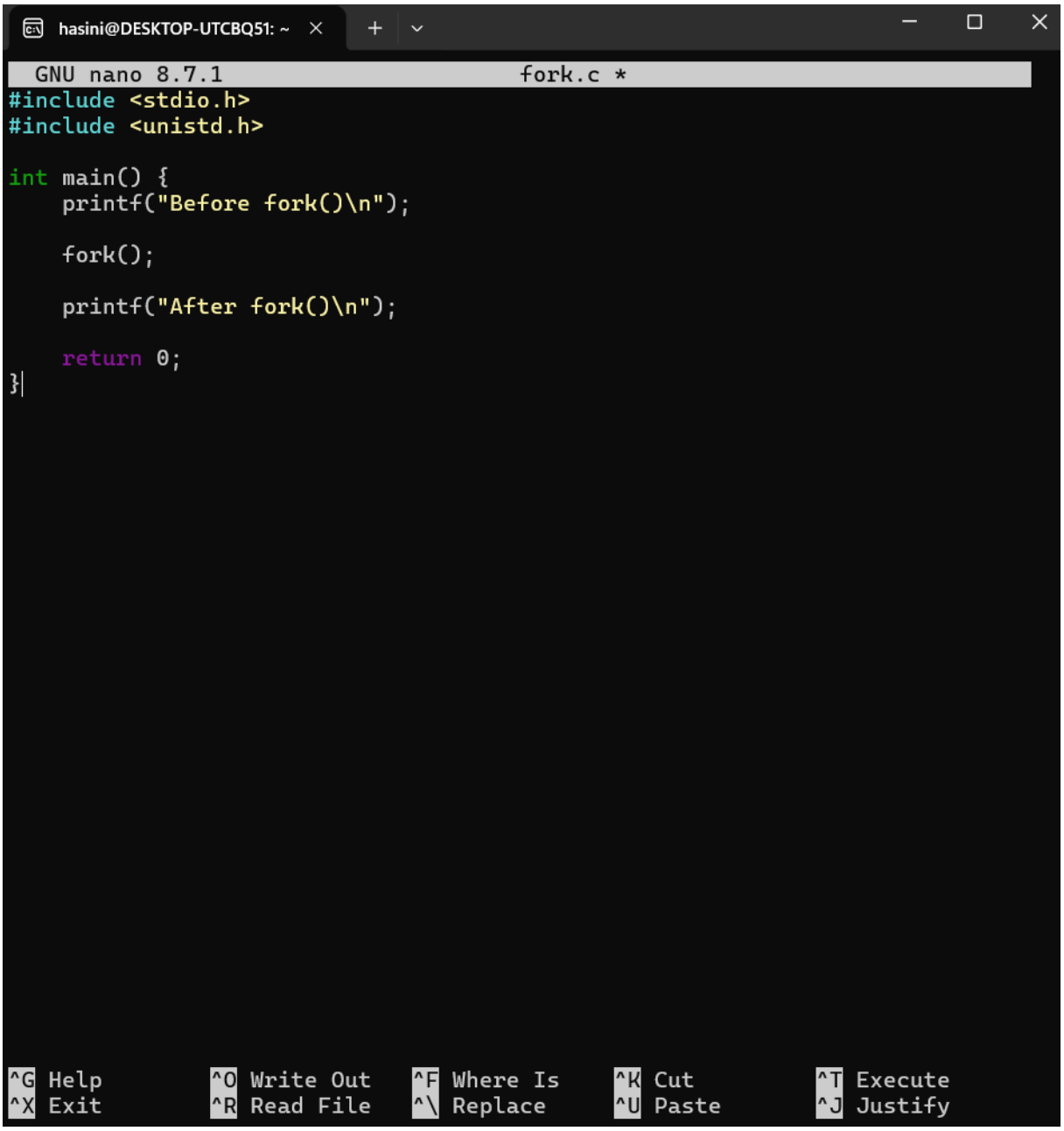
Name: J.chandra hasini
Roll_no: 2520030086

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

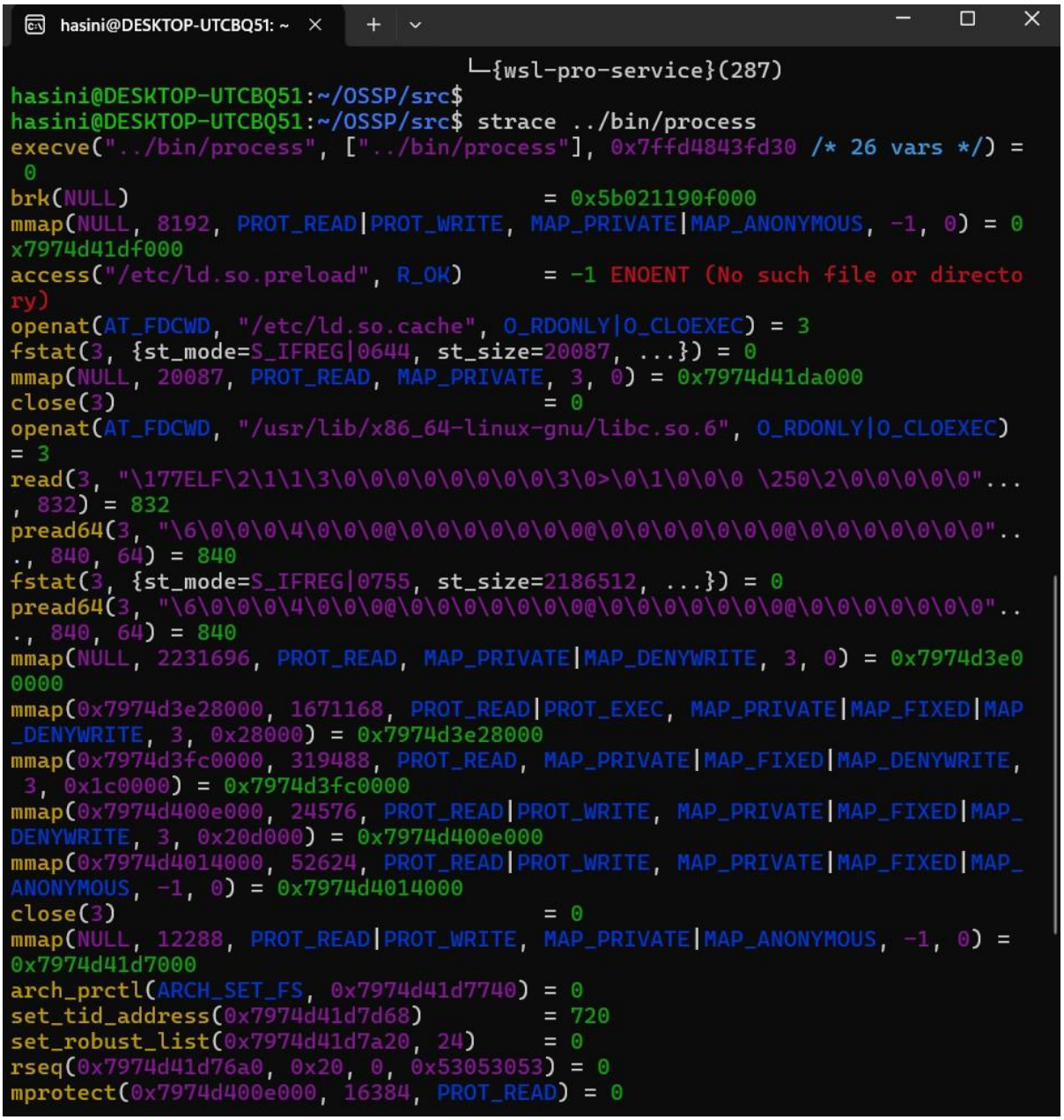


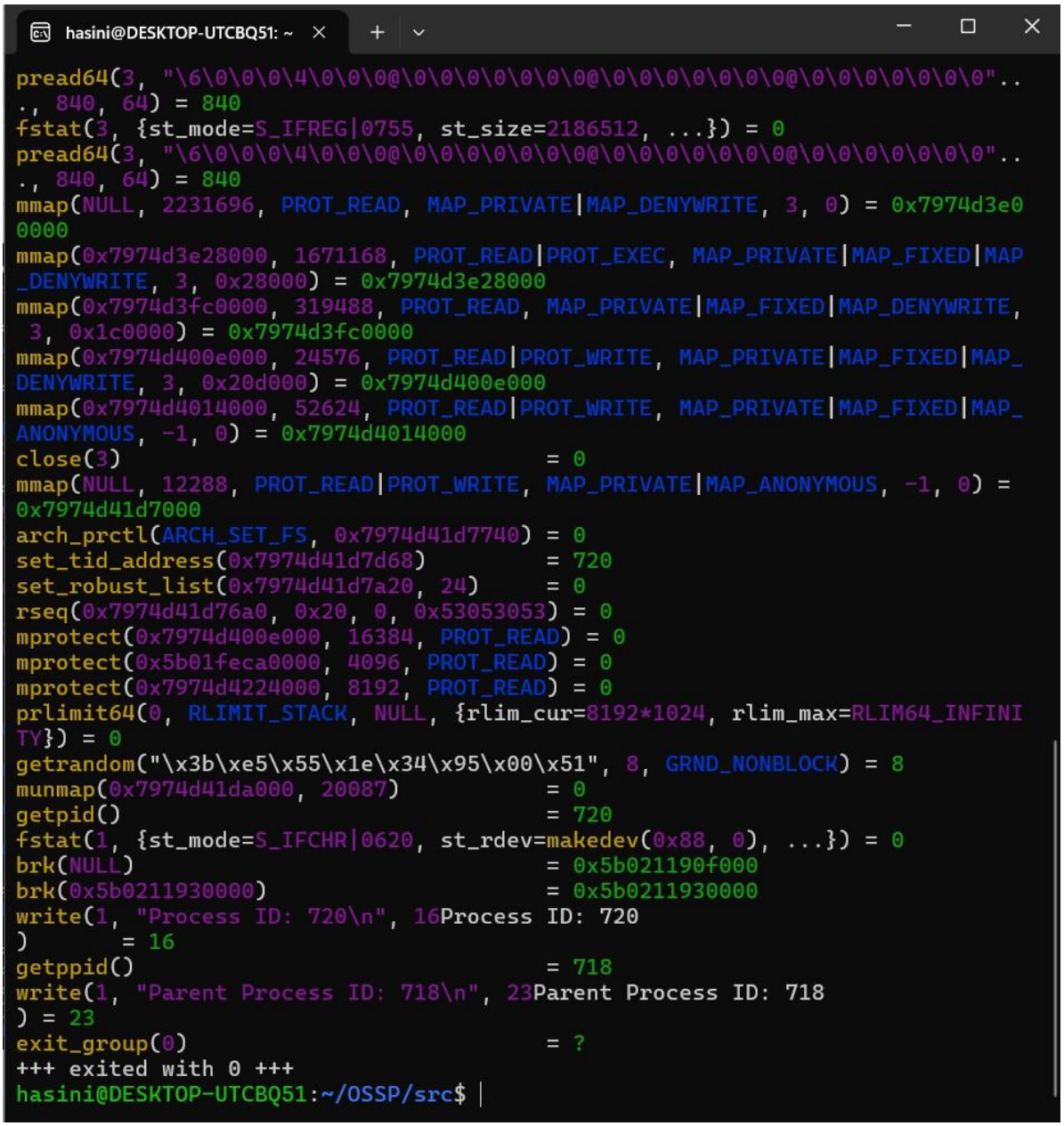



```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano fork.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc fork.c -o../bin/fork
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/fork
Before fork()
After fork()
After fork()
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano process.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc process.c -o ../bin/process
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/process
Process ID: 666
Parent Process ID: 338
```

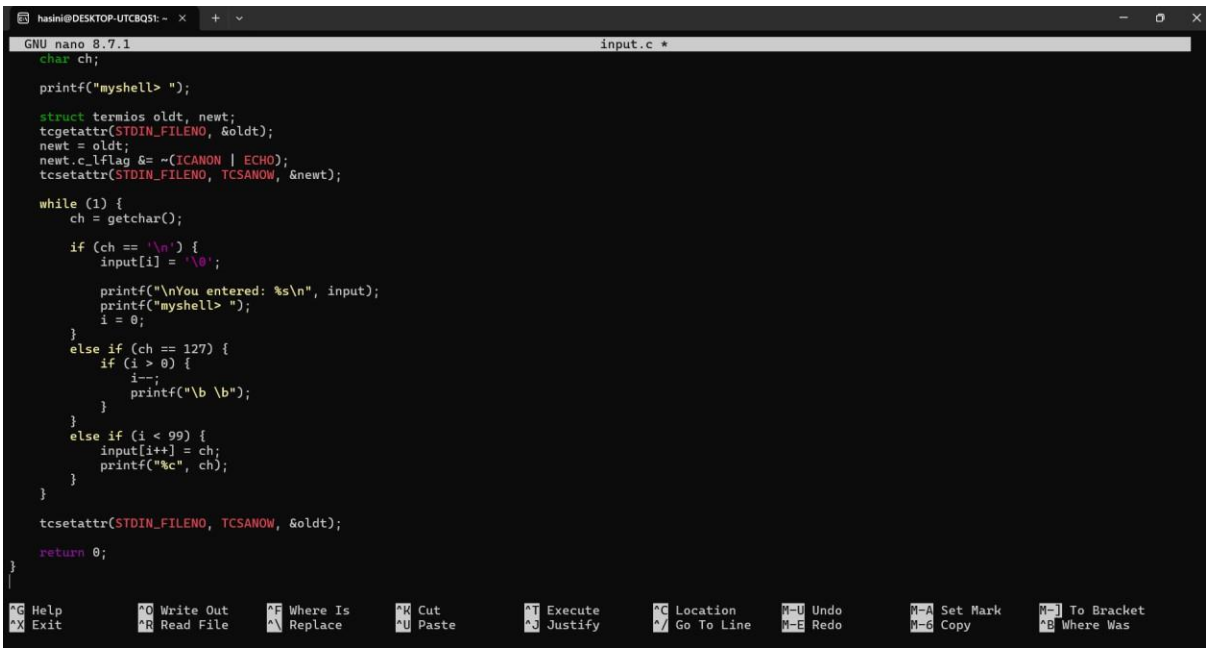

hasini@DESKTOP-UTCBQ51:~/OSSP/src$ pstree -p
systemd(1)─┬─agetty(207)
├─chronyd-starter(110)───chronyd(219)───chronyd(241)
├─cron(111)
├─dbus-daemon(112)
├─init-systemd(Ub(2)─┬─SessionLeader(335)───Relay(338)(337)───ba+
├─init(7)───{init}(8)
├─login(340)───bash(431)
└─{init-systemd(Ub}(9)
├─networkd-dispat(116)
├─rsyslogd(189)─┬─{rsyslogd}(210)
├─{rsyslogd}(211)
└─{rsyslogd}(212)
├─systemd(391)───(sd-pam)(393)
├─systemd-journal(50)
├─systemd-logind(129)
├─systemd-resolve(82)
├─systemd-udevd(91)─┬─(udev-worker)(695)
└─(udev-worker)(696)
├─unattended-upgr(220)───{unattended-upgr}(300)
└─wsl-pro-service(139)─┬─{wsl-pro-service}(224)
├─{wsl-pro-service}(225)
├─{wsl-pro-service}(226)
├─{wsl-pro-service}(227)
├─{wsl-pro-service}(231)
├─{wsl-pro-service}(232)
├─{wsl-pro-service}(236)
└─{wsl-pro-service}(287)

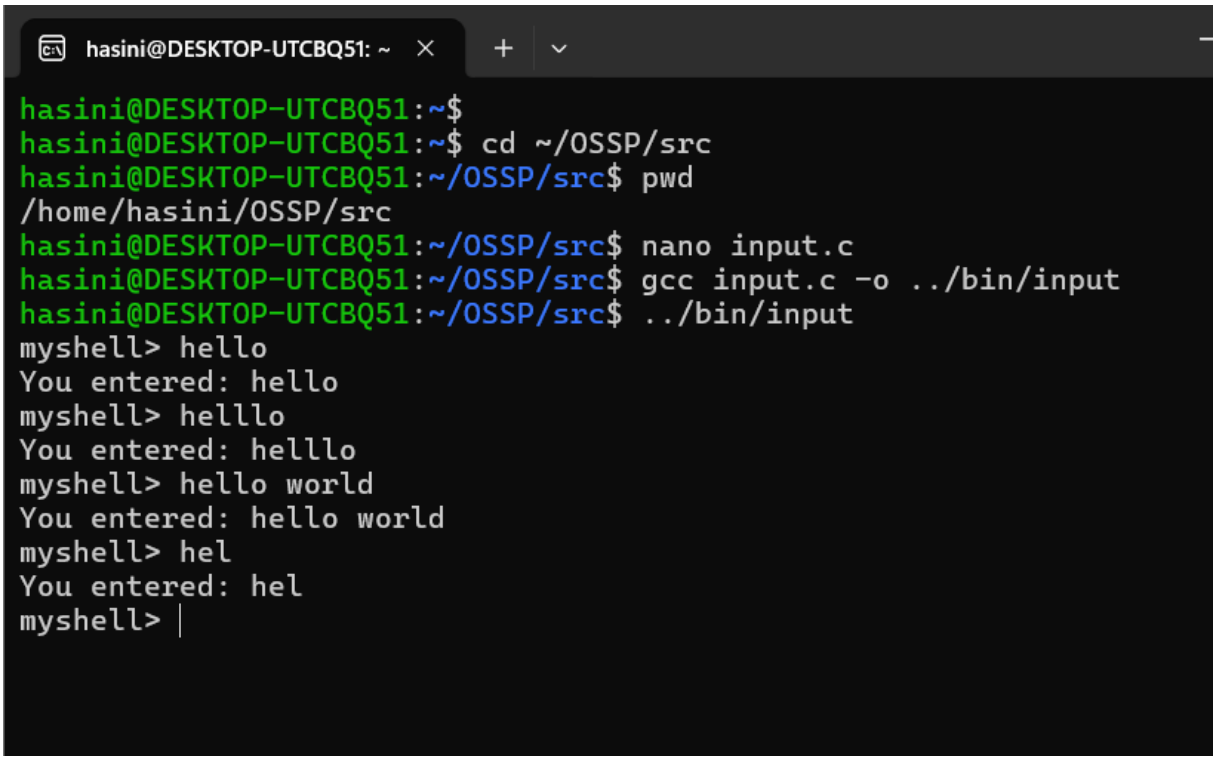

hasini@DESKTOP-UTCBQ51: ~
└─{wsl-pro-service}(287)
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ strace ../bin/process
execve("../bin/process", ["../bin/process"], 0x7ffd4843fd30 /* 26 vars */) = 0
brk(NULL) = 0x5b021190f000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7974d41df000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7974d41da000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7974d3e00000
mmap(0x7974d3e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7974d3e28000
mmap(0x7974d3fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x7974d3fc0000
mmap(0x7974d400e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x7974d400e000
mmap(0x7974d4014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7974d4014000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7974d41d7000
arch_prctl(ARCH_SET_FS, 0x7974d41d7740) = 0
set_tid_address(0x7974d41d7d68) = 720
set_robust_list(0x7974d41d7a20, 24) = 0
rseq(0x7974d41d76a0, 0x20, 0, 0x53053053) = 0
mprotect(0x7974d400e000, 16384, PROT_READ) = 0


hasini@DESKTOP-UTCBQ51: ~
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..
., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..
., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7974d3e0
0000
mmap(0x7974d3e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP
_DENYWRITE, 3, 0x28000) = 0x7974d3e28000
mmap(0x7974d3fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x1c0000) = 0x7974d3fc0000
mmap(0x7974d400e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x20d000) = 0x7974d400e000
mmap(0x7974d4014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_
ANONYMOUS, -1, 0) = 0x7974d4014000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7974d41d7000
arch_prctl(ARCH_SET_FS, 0x7974d41d7740) = 0
set_tid_address(0x7974d41d7d68) = 720
set_robust_list(0x7974d41d7a20, 24) = 0
rseq(0x7974d41d76a0, 0x20, 0, 0x53053053) = 0
mprotect(0x7974d400e000, 16384, PROT_READ) = 0
mprotect(0x5b01feca0000, 4096, PROT_READ) = 0
mprotect(0x7974d4224000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINI
TY}) = 0
getrandom("\x3b\xe5\x55\x1e\x34\x95\x00\x51", 8, GRND_NONBLOCK) = 8
munmap(0x7974d41da000, 20087) = 0
getpid() = 720
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
brk(NULL) = 0x5b021190f000
brk(0x5b0211930000) = 0x5b0211930000
write(1, "Process ID: 720\n", 16Process ID: 720
) = 16
getppid() = 718
write(1, "Parent Process ID: 718\n", 23Parent Process ID: 718
) = 23
exit_group(0) = ?
+++ exited with 0 +++
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

2.


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 input.c *
#include <stdio.h>
#include <termios.h>
#include <unistd.h>
int main() {
char input[100];
int i = 0;
char ch;
printf("myshell> ");
struct termios oldt, newt;
tcgetattr(STDIN_FILENO, &oldt);
newt = oldt;
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);
while (1) {
ch = getchar();
if (ch == '\n') {
input[i] = '\0';
printf("\nYou entered: %s\n", input);
printf("myshell> ");
i = 0;
}
else if (ch == 127) {
if (i > 0) {
i--;
printf("\b \b");
}
}
else if (i < 99) {
input[i++] = ch;
printf("%c", ch);
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 input.c *
char ch;
printf("myshell> ");
struct termios oldt, newt;
tcgetattr(STDIN_FILENO, &oldt);
newt = oldt;
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);
while (1) {
ch = getchar();
if (ch == '\n') {
input[i] = '\0';
printf("\nYou entered: %s\n", input);
printf("myshell> ");
i = 0;
}
else if (ch == 127) {
if (i > 0) {
i--;
printf("\b \b");
}
}
else if (i < 99) {
input[i++] = ch;
printf("%c", ch);
}
}
tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was

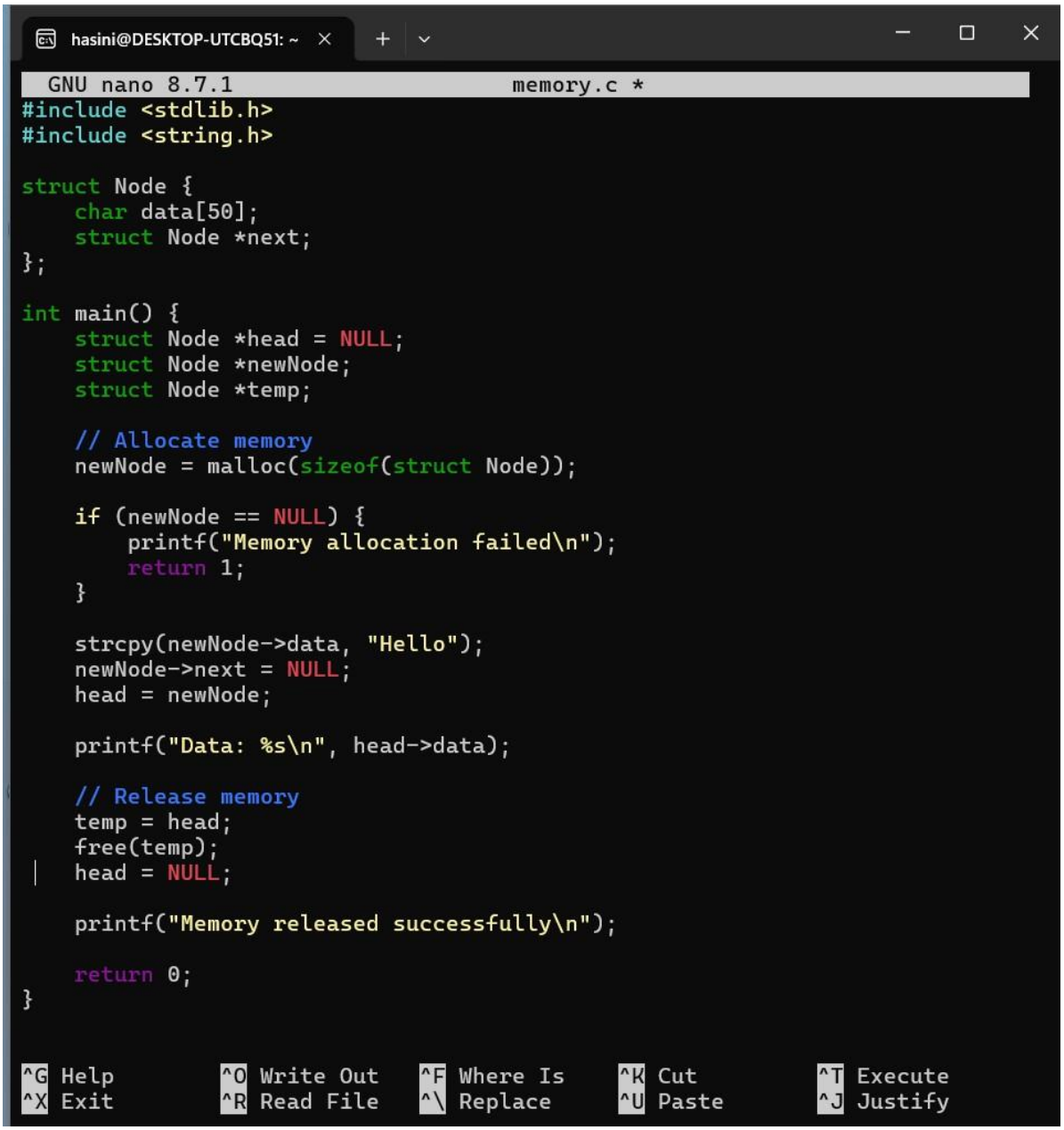

hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ pwd
/home/hasini/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano input.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc input.c -o ../bin/input
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/input
myshell> hello
You entered: hello
myshell> helllo
You entered: helllo
myshell> hello world
You entered: hello world
myshell> hel
You entered: hel
myshell>

3.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ pwd
/home/hasini/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano history.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 history.c *
#define MAX_HISTORY 10
char *history[MAX_HISTORY];
int history_count = 0;
int history_pos = 0;
void add_history(char *command) {
if (strlen(command) == 0)
return;
if (history_count < MAX_HISTORY) {
history[history_count] = malloc(strlen(command) + 1);
strcpy(history[history_count], command);
history_count++;
}
}
void show_history_command() {
if (history_pos >= 0 && history_pos < history_count)
printf("\r\033[Kmyshell> %s", history[history_pos]);
}
int main() {
char input[100];
int i = 0;
char ch;
struct termios oldt, newt;
tcgetattr(STDIN_FILENO, &oldt);
newt = oldt;
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);
while (1) {
printf("\nmyshell> ");
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 history.c *
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);
while (1) {
printf("\nmyshell> ");
fflush(stdout);
i = 0;
while (1) {
ch = getchar();
if (ch == '\n') {
input[i] = '\0';
if (strcmp(input, "exit") == 0)
break;
printf("\nYou entered: %s\n", input);
add_history(input);
history_pos = history_count;
break;
}
else if (ch == 127) {
if (i > 0) {
i--;
printf("\b \b");
fflush(stdout);
}
}
else if (ch == 27) {
char ch1 = getchar();
char ch2 = getchar();
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 memory.c *
#include <stdlib.h>
#include <string.h>
struct Node {
char data[50];
struct Node *next;
};
int main() {
struct Node *head = NULL;
struct Node *newNode;
struct Node *temp;
// Allocate memory
newNode = malloc(sizeof(struct Node));
if (newNode == NULL) {
printf("Memory allocation failed\n");
return 1;
}
strcpy(newNode->data, "Hello");
newNode->next = NULL;
head = newNode;
printf("Data: %s\n", head->data);
// Release memory
temp = head;
free(temp);
head = NULL;
printf("Memory released successfully\n");
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify



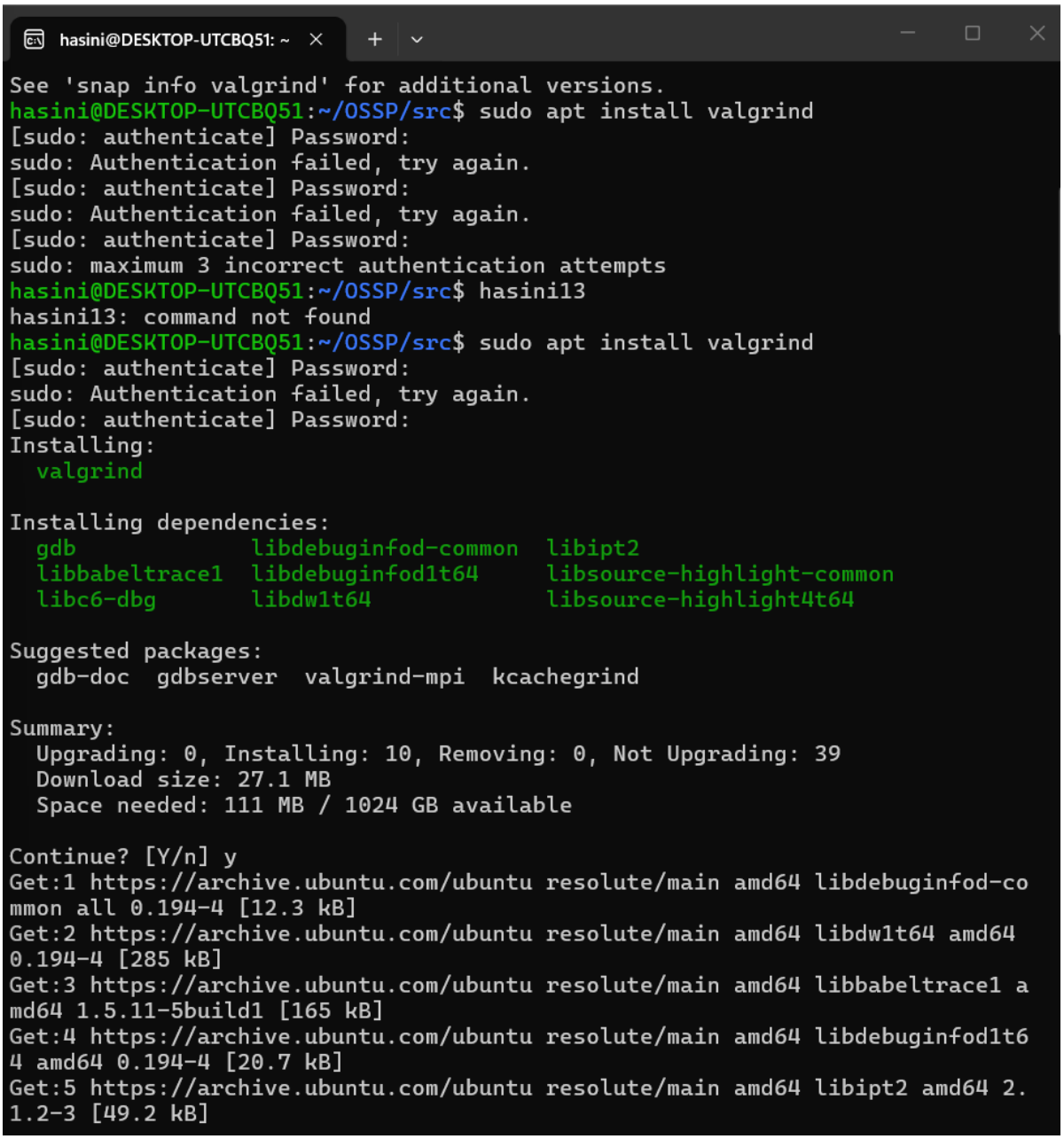

```
myshell> hello world
You entered: hello world

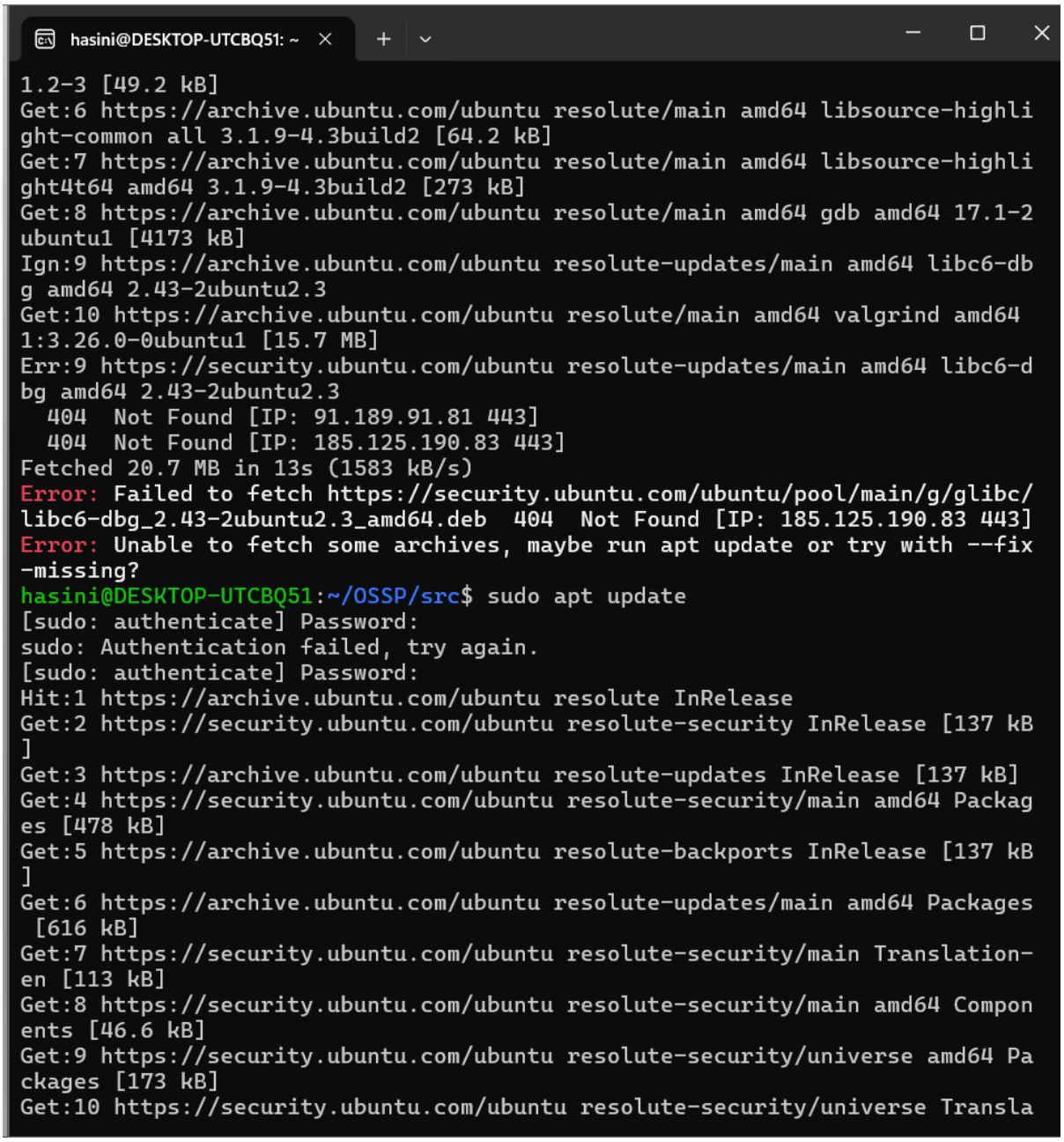
myshell> hello world
You entered:

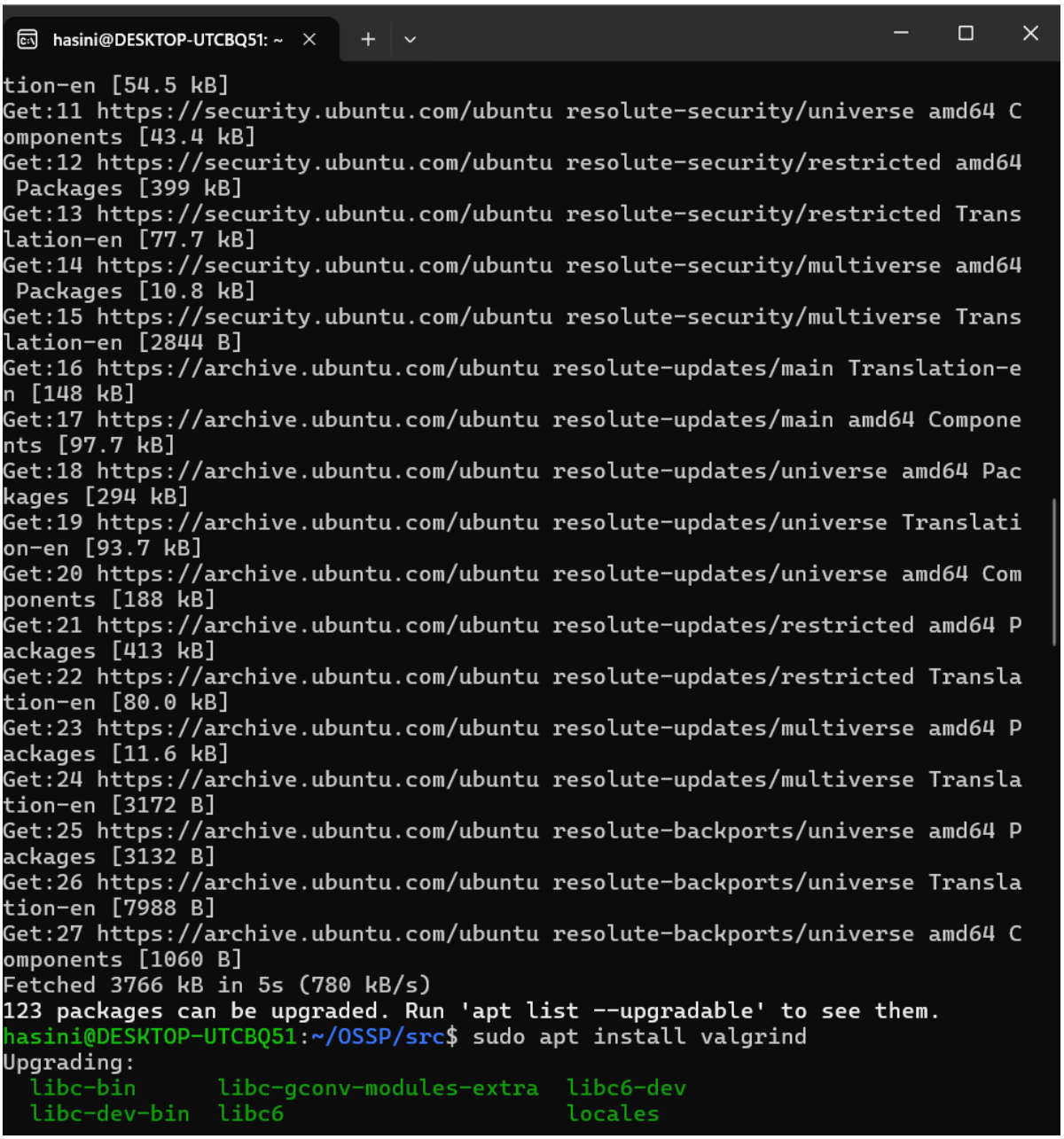
myshell> test
You entered: test

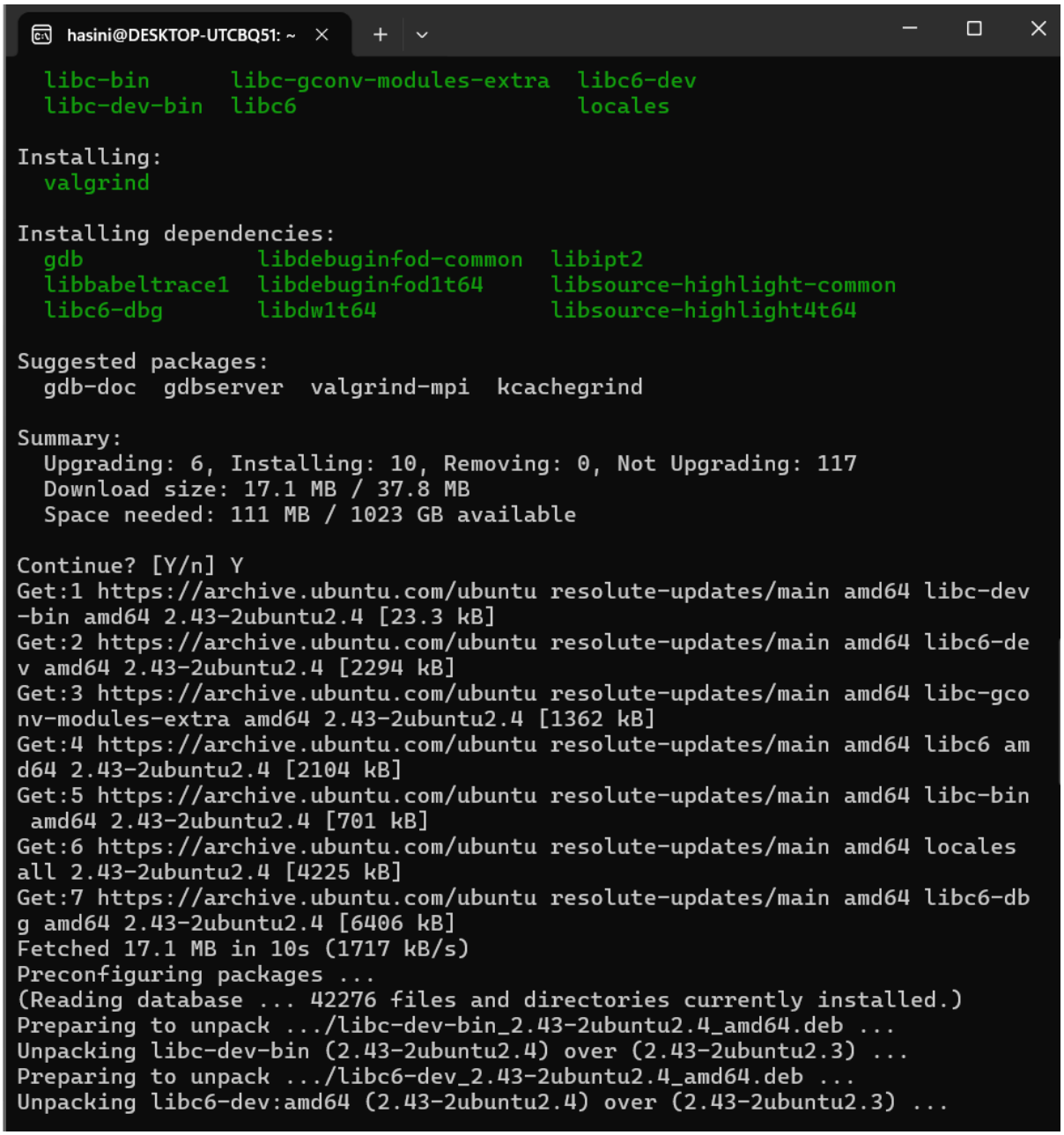
myshell> exithasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano memory.c
```

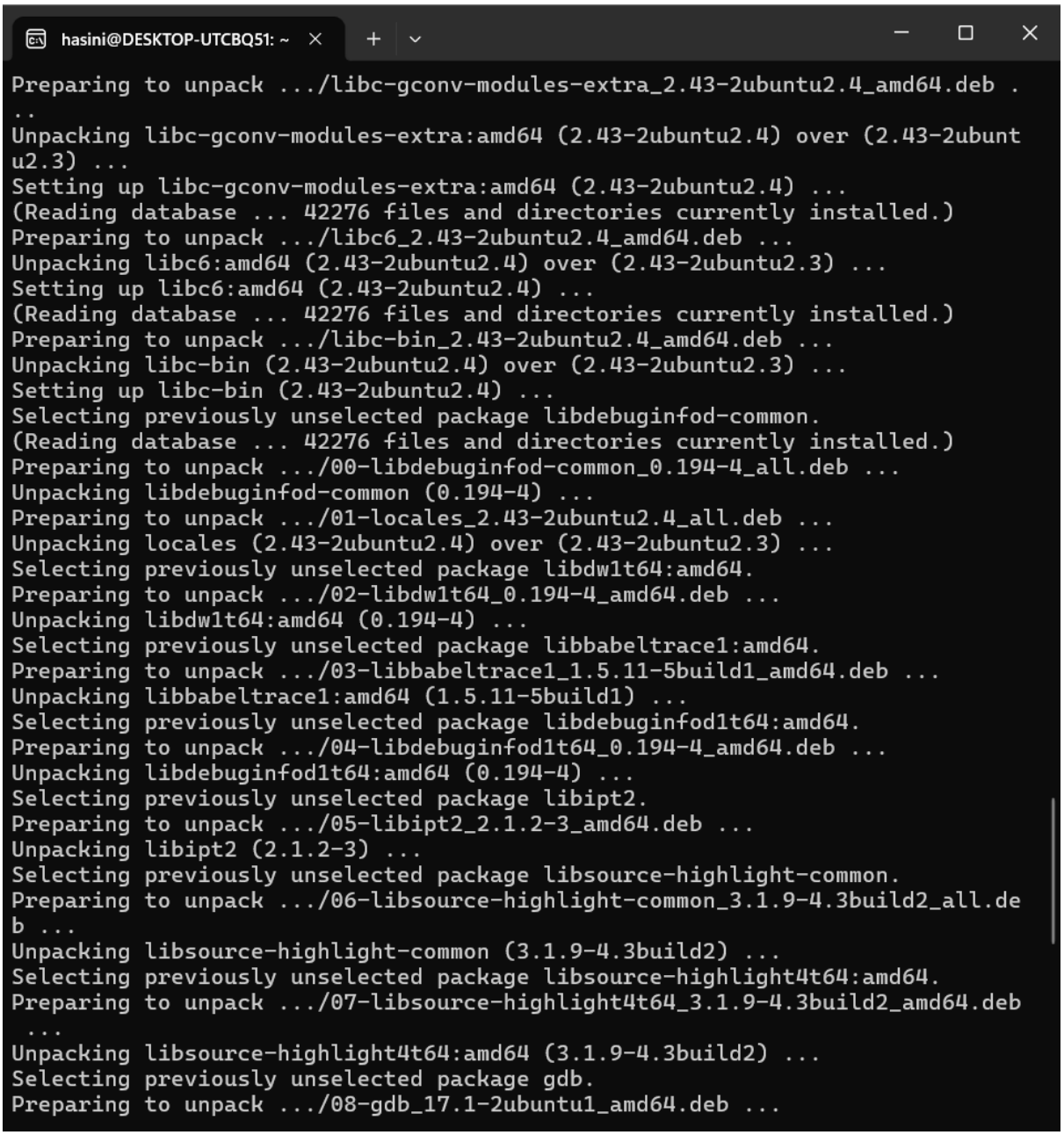
```
myshell> exithasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano memory.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc memory.c -o ../bin/memory
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/memory
Data: Hello
Memory released successfully
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind  # version 3.26.0, or
sudo apt  install valgrind  # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ hasini13
hasini13: command not found
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

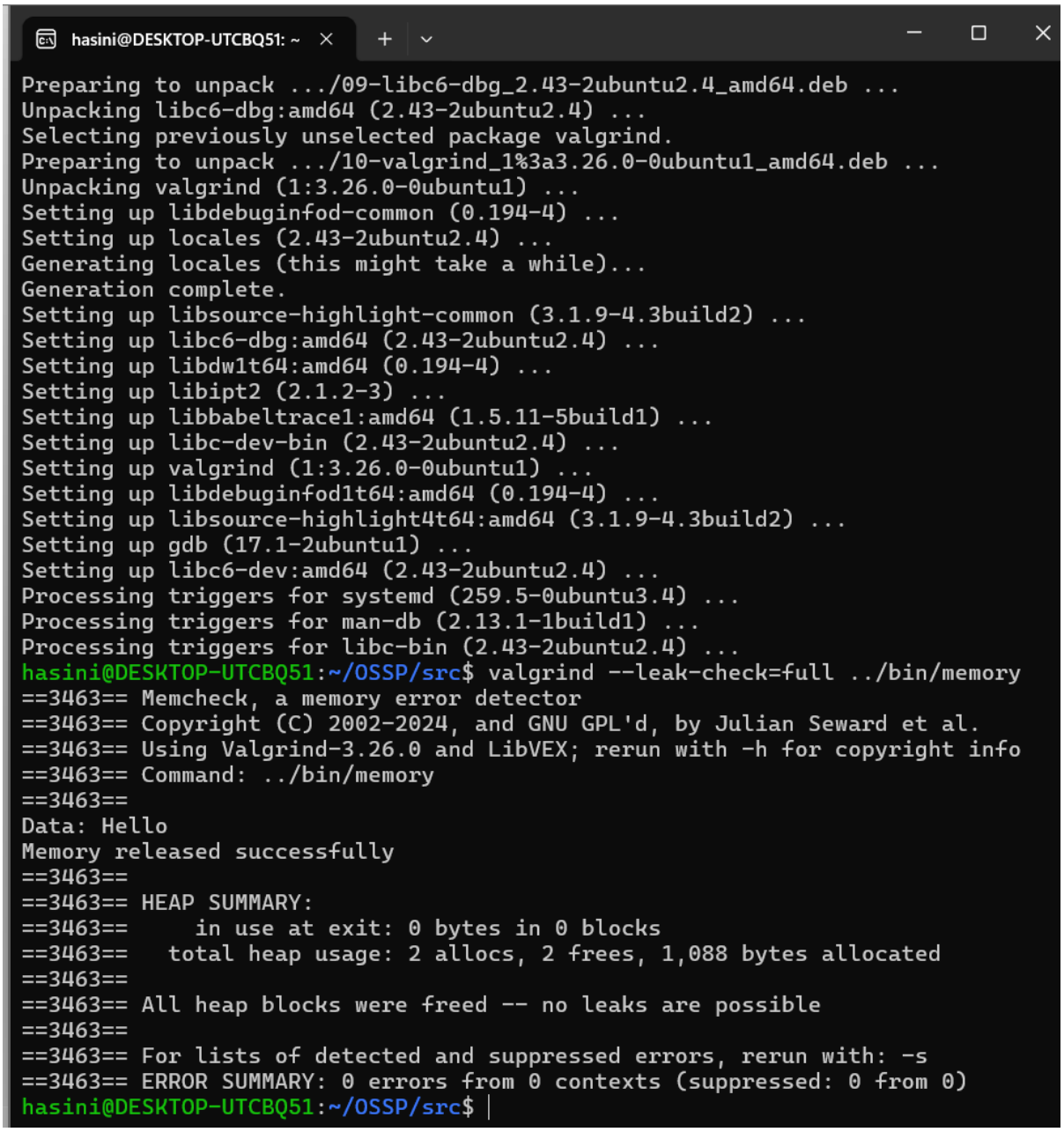

hasini@DESKTOP-UTCBQ51: ~
See 'snap info valgrind' for additional versions.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ hasini13
hasini13: command not found
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Installing:
valgrind
Installing dependencies:
gdb libdebuginfod-common libipt2
libbabeltrace1 libdebuginfod1t64 libsource-highlight-common
libc6-dbg libdw1t64 libsource-highlight4t64
Suggested packages:
gdb-doc gdbserver valgrind-mpi kcachegrind
Summary:
Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 39
Download size: 27.1 MB
Space needed: 111 MB / 1024 GB available
Continue? [Y/n] y
Get:1 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 https://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 https://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]


hasini@DESKTOP-UTCBQ51: ~
1.2-3 [49.2 kB]
Get:6 https://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 https://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:8 https://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]
Ign:9 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3
Get:10 https://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]
Err:9 https://security.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3
404 Not Found [IP: 91.189.91.81 443]
404 Not Found [IP: 185.125.190.83 443]
Fetched 20.7 MB in 13s (1583 kB/s)
Error: Failed to fetch https://security.ubuntu.com/ubuntu/pool/main/g/glibc/libc6-dbg_2.43-2ubuntu2.3_amd64.deb 404 Not Found [IP: 185.125.190.83 443]
Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hit:1 https://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 https://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 https://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 https://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [478 kB]
Get:5 https://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
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hasini@DESKTOP-UTCBQ51: ~
tion-en [54.5 kB]
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Get:15 https://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2844 B]
Get:16 https://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [148 kB]
Get:17 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.7 kB]
Get:18 https://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [294 kB]
Get:19 https://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [93.7 kB]
Get:20 https://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:21 https://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [413 kB]
Get:22 https://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [80.0 kB]
Get:23 https://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:24 https://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3172 B]
Get:25 https://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3132 B]
Get:26 https://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:27 https://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1060 B]
Fetched 3766 kB in 5s (780 kB/s)
123 packages can be upgraded. Run 'apt list --upgradable' to see them.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ sudo apt install valgrind
Upgrading:
libc-bin libc-gconv-modules-extra libc6-dev
libc-dev-bin libc6 locales


hasini@DESKTOP-UTCBQ51: ~
libc-bin libc-gconv-modules-extra libc6-dev
libc-dev-bin libc6 locales
Installing:
valgrind
Installing dependencies:
gdb libdebuginfod-common libipt2
libbabeltrace1 libdebuginfod1t64 libsource-highlight-common
libc6-dbg libdw1t64 libsource-highlight4t64
Suggested packages:
gdb-doc gdbserver valgrind-mpi kcachegrind
Summary:
Upgrading: 6, Installing: 10, Removing: 0, Not Upgrading: 117
Download size: 17.1 MB / 37.8 MB
Space needed: 111 MB / 1023 GB available
Continue? [Y/n] Y
Get:1 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-dev-bin amd64 2.43-2ubuntu2.4 [23.3 kB]
Get:2 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dev amd64 2.43-2ubuntu2.4 [2294 kB]
Get:3 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-gconv-modules-extra amd64 2.43-2ubuntu2.4 [1362 kB]
Get:4 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6 amd64 2.43-2ubuntu2.4 [2104 kB]
Get:5 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-bin amd64 2.43-2ubuntu2.4 [701 kB]
Get:6 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 locales all 2.43-2ubuntu2.4 [4225 kB]
Get:7 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.4 [6406 kB]
Fetched 17.1 MB in 10s (1717 kB/s)
Preconfiguring packages ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc-dev-bin_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-dev-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Preparing to unpack .../libc6-dev_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dev:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...


hasini@DESKTOP-UTCBQ51: ~
Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc6_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc6:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-bin (2.43-2ubuntu2.4) ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Preparing to unpack .../01-locales_2.43-2ubuntu2.4_all.deb ...
Unpacking locales (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../02-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../03-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_17.1-2ubuntu1_amd64.deb ...


hasini@DESKTOP-UTCBQ51: ~
Preparing to unpack .../09-libc6-dbg_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up locales (2.43-2ubuntu2.4) ...
Generating locales (this might take a while)...
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.4) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
==3463== Memcheck, a memory error detector
==3463== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3463== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3463== Command: ../bin/memory
==3463==
Data: Hello
Memory released successfully
==3463==
==3463== HEAP SUMMARY:
==3463==   in use at exit: 0 bytes in 0 blocks
==3463==   total heap usage: 2 allocs, 2 frees, 1,088 bytes allocated
==3463==
==3463== All heap blocks were freed -- no leaks are possible
==3463==
==3463== For lists of detected and suppressed errors, rerun with: -s
==3463== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

4.

```
hasini@DESKTOP-UTCBQ51:~$ nano tokenize.c
hasini@DESKTOP-UTCBQ51:~$ gcc tokenize.c -o ../bin/tokenize
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../bin/tokenize: N
o such file or directory
collect2: error: ld returned 1 exit status
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ mv ~/tokenize.c .
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ pwd
/home/hasini/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc tokenize.c -o ../bin/tokenize
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/tokenize
Enter a command: ls -l file.txt
Tokens:
ls
-l
file.txt
```

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                    tokenize.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char *token;

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    token = strtok(input, " \n");

    printf("Tokens:\n");

    while (token != NULL) {
        printf("%s\n", token);
        token = strtok(NULL, " \n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano parse.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc parse.c -o ../bin/parse
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/parse
Enter command: ls file.txt
Command: ls
Argument: file.txt
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                    parse.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char *command;
    char *argument;

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    command = strtok(input, " \n");
    argument = strtok(NULL, " \n");

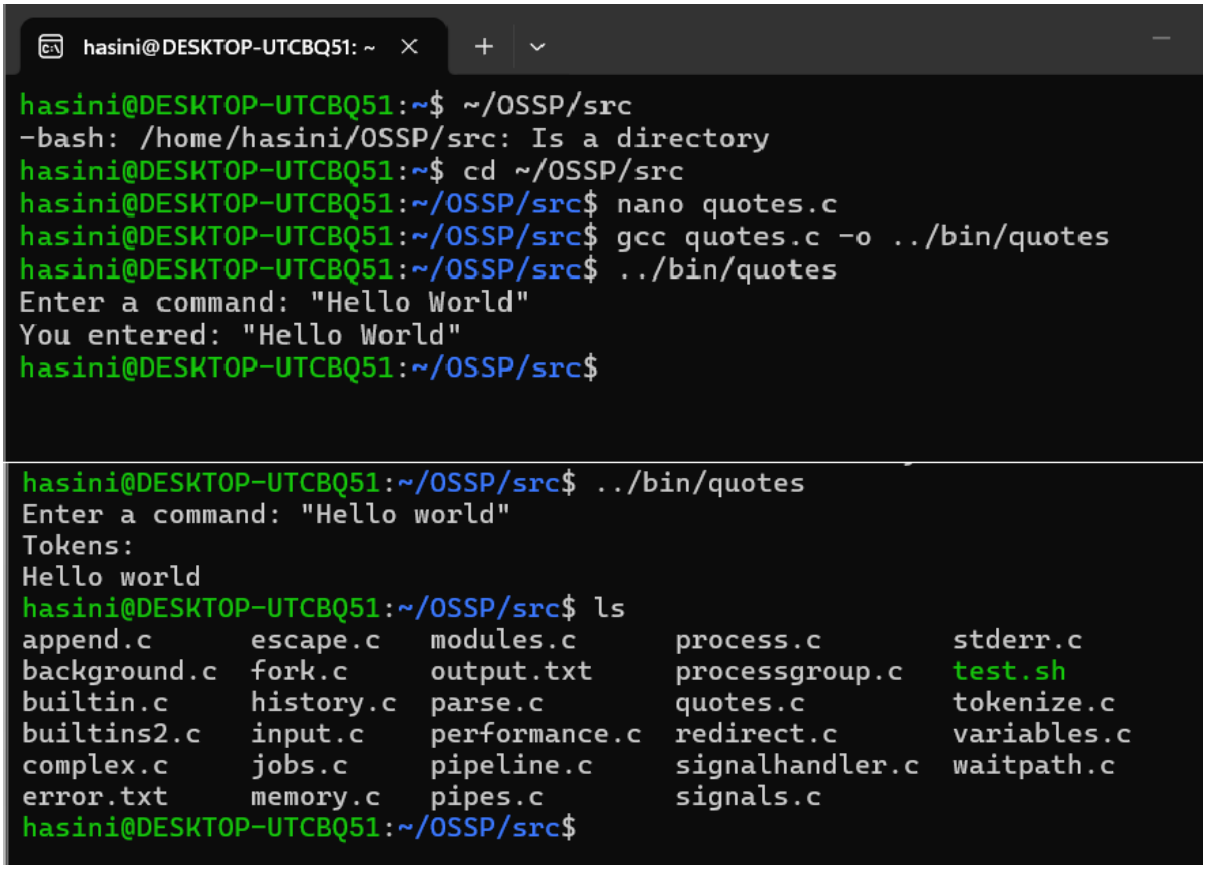
    printf("Command: %s\n", command);

    if (argument != NULL) {
        printf("Argument: %s\n", argument);
    }

    return 0;
}

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify
```

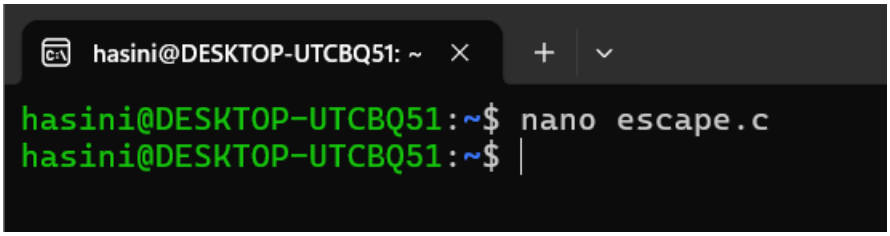
5.

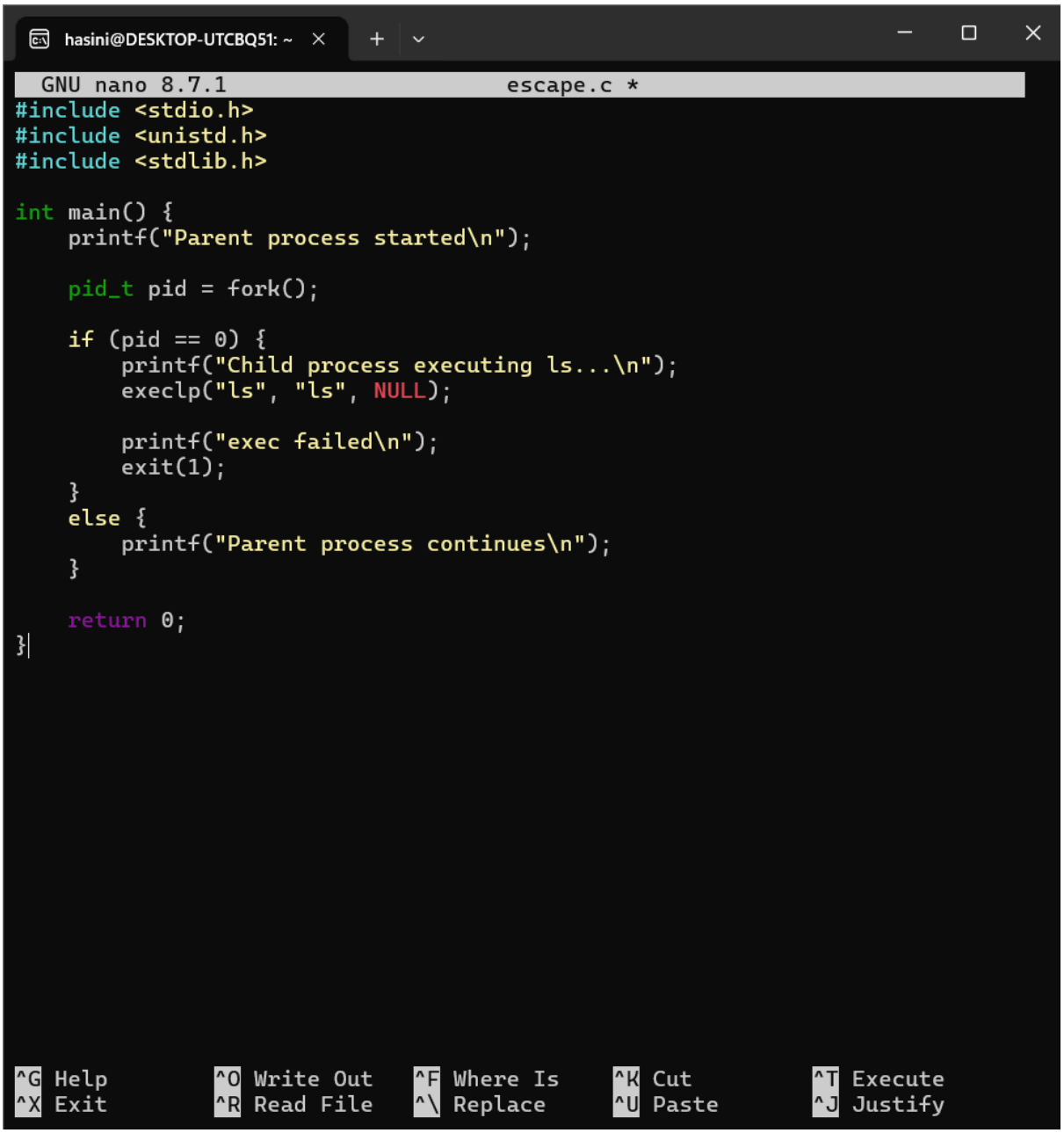

hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano quotes.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc quotes.c -o ../bin/quotes
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/quotes
Enter a command: "Hello World"
You entered: "Hello World"
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/quotes
Enter a command: "Hello world"
Tokens:
Hello world
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ls
append.c escape.c modules.c process.c stderr.c
background.c fork.c output.txt processgroup.c test.sh
builtin.c history.c parse.c quotes.c tokenize.c
builtins2.c input.c performance.c redirect.c variables.c
complex.c jobs.c pipeline.c signalhandler.c waitpath.c
error.txt memory.c pipes.c signals.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 quotes.c *
#include <stdio.h>
#include <string.h>
int main()
{
char input[200];
char token[100];
int i = 0;
int j = 0;
char quote = 0;
printf("Enter a command: ");
fgets(input, sizeof(input), stdin);
printf("Tokens:\n");
while (input[i] != '\0' && input[i] != '\n')
{
if (input[i] == '"' || input[i] == '\'')
{
if (quote == 0)
{
quote = input[i];
}
else if (quote == input[i])
{
quote = 0;
}
else
{
token[j++] = input[i];
}
}
else if (input[i] == ' ' && quote == 0)
{
if (j > 0)
{
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 quotes.c *
quote = input[i];
}
else if (quote == input[i])
{
quote = 0;
}
else
{
token[j++] = input[i];
}
}
else if (input[i] == ' ' && quote == 0)
{
if (j > 0)
{
token[j] = '\0';
printf("%s\n", token);
j = 0;
}
}
else
{
token[j++] = input[i];
}
i++;
}
if (j > 0)
{
token[j] = '\0';
printf("%s\n", token);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

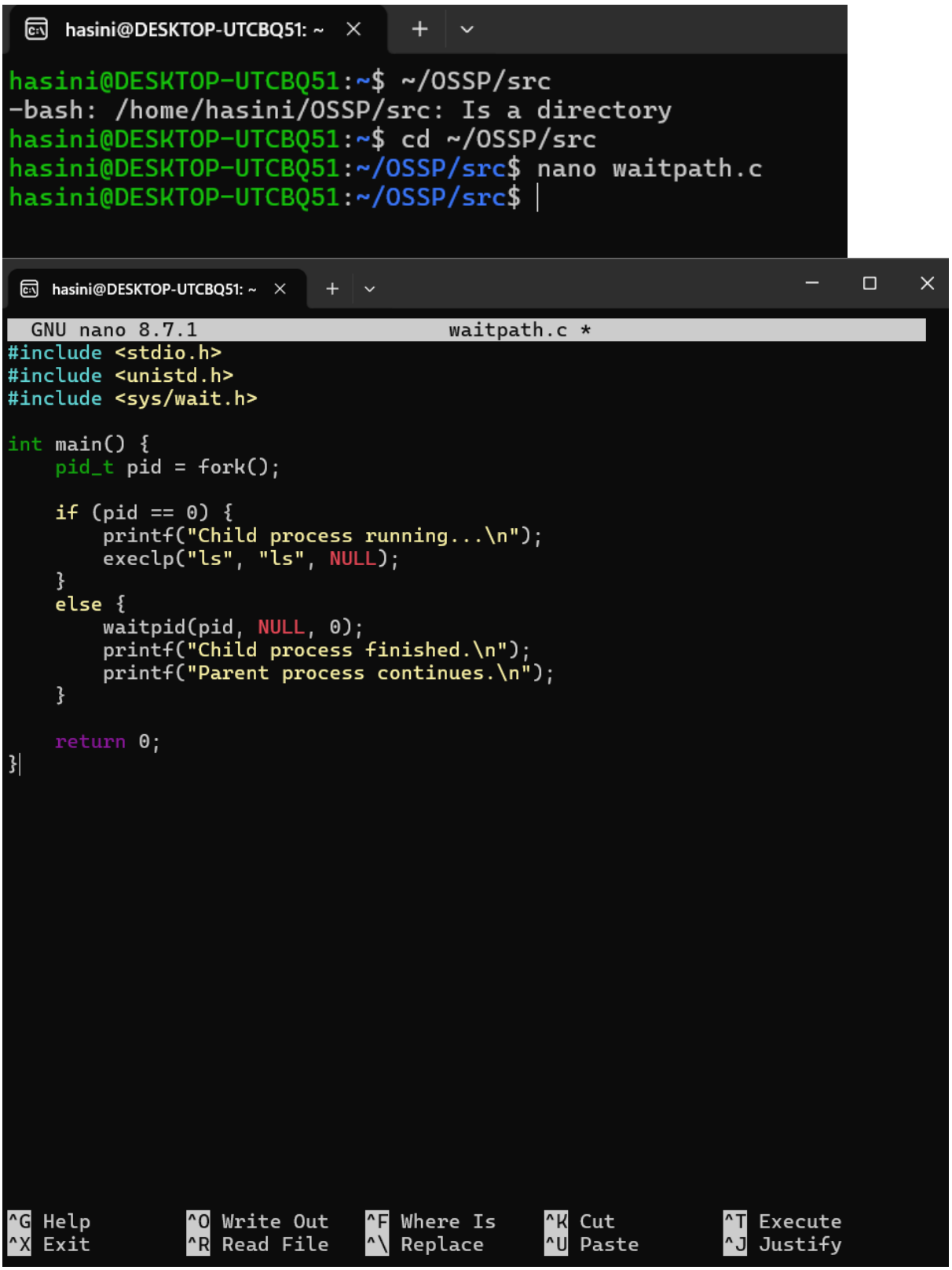
6.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ nano escape.c
hasini@DESKTOP-UTCBQ51:~$




```
hasini@DESKTOP-UTCBQ51:~$ gcc escape.c -o ../bin/escape
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../bin/escape: No
such file or directory
collect2: error: ld returned 1 exit status
hasini@DESKTOP-UTCBQ51:~$ mv ~/escape.c ~/OSSP/src/
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc escape.c -o ../bin/escape
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/escape
Parent process started
Parent process continues
Child process executing ls...
escape.c  history.c  memory.c  process.c  tokenize.c
fork.c    input.c    parse.c   quotes.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

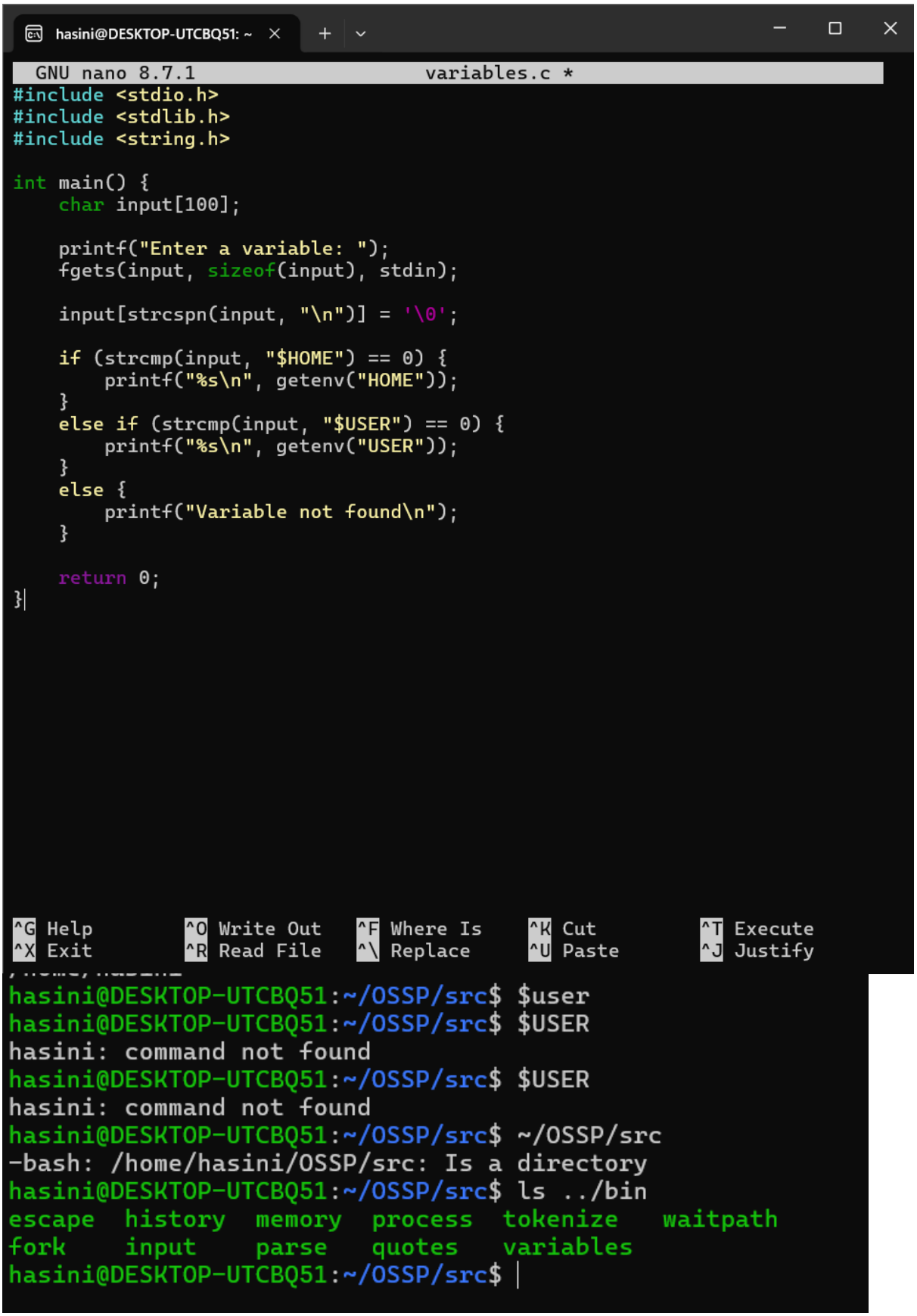
7.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano waitpath.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 waitpath.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
pid_t pid = fork();
if (pid == 0) {
printf("Child process running...\n");
execlp("ls", "ls", NULL);
}
else {
waitpid(pid, NULL, 0);
printf("Child process finished.\n");
printf("Parent process continues.\n");
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano waitpath.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc waitpath.c -o ../bin/waitpath
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/waitpath
Child process running...
escape.c  history.c  memory.c  process.c  tokenize.c
fork.c    input.c    parse.c   quotes.c   waitpath.c
Child process finished.
Parent process continues.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

8.

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano variables.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc variables.c -o ../bin/variables
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/variables
Enter a variable: $HOME
/home/hasini
```


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 variables.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main() {
char input[100];
printf("Enter a variable: ");
fgets(input, sizeof(input), stdin);
input[strcspn(input, "\n")] = '\0';
if (strcmp(input, "$HOME") == 0) {
printf("%s\n", getenv("HOME"));
}
else if (strcmp(input, "$USER") == 0) {
printf("%s\n", getenv("USER"));
}
else {
printf("Variable not found\n");
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ $user
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ $USER
hasini: command not found
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ $USER
hasini: command not found
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ls ../bin
escape history memory process tokenize waitpath
fork input parse quotes variables
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

9.


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 builtin.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
int main() {
char command[100];
while (1) {
printf("myshell> ");
fgets(command, sizeof(command), stdin);
command[strcspn(command, "\n")] = '\0';
if (strcmp(command, "exit") == 0) {
break;
}
if (strncmp(command, "cd ", 3) == 0) {
if (chdir(command + 3) == 0)
printf("Directory changed successfully\n");
else
printf("Directory change failed\n");
}
else {
printf("Unknown command\n");
}
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano builtin.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc builtin.c -o ../bin/builtin
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/builtin
myshell> cd..
Unknown command
myshell> cd..
Unknown command
myshell> cd ..
Directory changed successfully
myshell> cd src
Directory changed successfully
myshell> exit
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

10.

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src~/OSSP/src
-bash: /home/hasini/OSSP/src~/OSSP/src: No such file or directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
```

```
 GNU nano 8.7.1                    builtins2.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char command[100];

    while (1) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "pwd") == 0) {
            char path[200];
            getcwd(path, sizeof(path));
            printf("%s\n", path);
        }
        else if (strcmp(command, "exit") == 0) {
            break;
        }
        else if (strncmp(command, "export ", 7) == 0) {
            char *value = command + 7;
            putenv(value);
            printf("Variable exported\n");
        }
        else {
            printf("Unknown command\n");
        }
    }

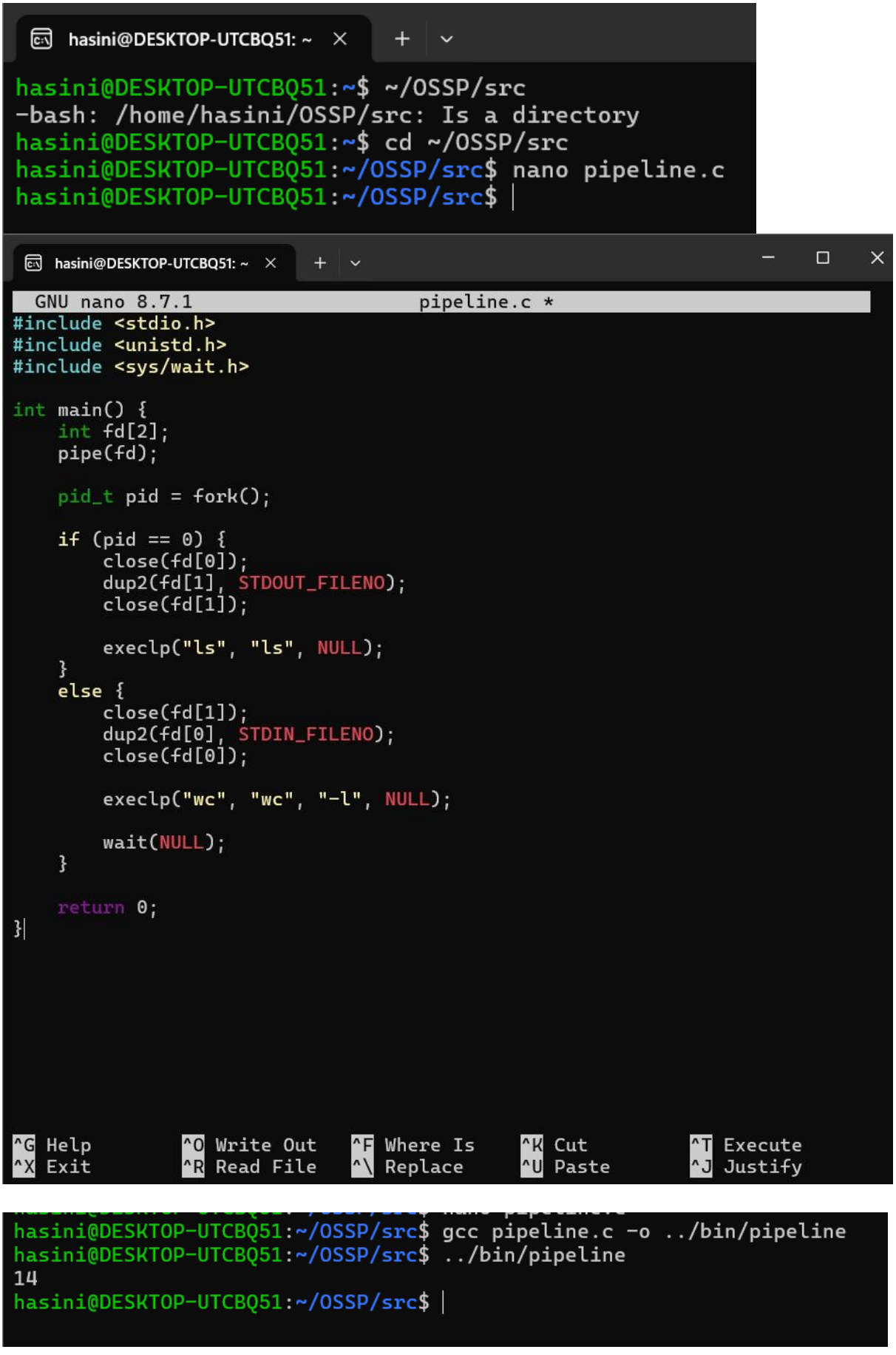
    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano builtins2.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc builtins2.c -o ../bin/builtins2
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/builtins2
myshell> pwd
/home/hasini/OSSP/src
myshell> export NAME=Hasini
Variable exported
myshell> exit
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

11.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano pipeline.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 pipeline.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
int fd[2];
pipe(fd);
pid_t pid = fork();
if (pid == 0) {
close(fd[0]);
dup2(fd[1], STDOUT_FILENO);
close(fd[1]);
execlp("ls", "ls", NULL);
}
else {
close(fd[1]);
dup2(fd[0], STDIN_FILENO);
close(fd[0]);
execlp("wc", "wc", "-l", NULL);
wait(NULL);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc pipeline.c -o ../bin/pipeline
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/pipeline
14
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

12.

```
hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc pipes.c -o ../bin/pipes
cc1: fatal error: pipes.c: No such file or directory
compilation terminated.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano pipes.c
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano pipes.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc pipes.c -o ../bin/pipes
```

```
hasini@DESKTOP-UTCBQ51: ~
 GNU nano 8.7.1                          pipes.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    char message[] = "Hello from child process";
    char buffer[100];

    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);
    }
    else {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Parent received: %s\n", buffer);

        close(fd[0]);
        wait(NULL);
    }

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc pipes.c -o ../bin/pipes
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/pipes
Parent received: Hello from child process
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

13.

```
hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano redirect.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

```
hasini@DESKTOP-UTCBQ51: ~
 GNU nano 8.7.1                    redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text is redirected to a file.\n");

    return 0;
}

^G Help      ^O Write Out   ^F Where Is   ^K Cut       ^T Execute
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc redirect.c -o ../bin/redirect
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/redirect
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ cat output.txt
This text is redirected to a file.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

14.

```
hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano append.c
```

```
hasini@DESKTOP-UTCBQ51: ~
 GNU nano 8.7.1                    append.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This line is appended.\n");

    return 0;
}
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc append.c -o ../bin/append
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/append
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ cat output.txt
This text is redirected to a file.
This line is appended.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano stderr.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc stderr.c -o ../bin/stderr
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc stderr.c -o ../bin/stderr
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/stderr
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ cat error.txt
This is an error message.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                    stderr.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        return 1;
    }

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "This is an error message.\n");

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

15.

```
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano complex.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

hasini@DESKTOP-UTCBQ51: ~

```
hasini@DESKTOP-UTCBQ51:~$
hasini@DESKTOP-UTCBQ51:~$ #include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char input[200];

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        return 0;

    char *command1 = strtok(input, "|");
    char *command2 = strtok(NULL, "|");

    if (command2 != NULL)
    {
        int pipefd[2];
        pipe(pipefd);

        pid_t pid1 = fork();

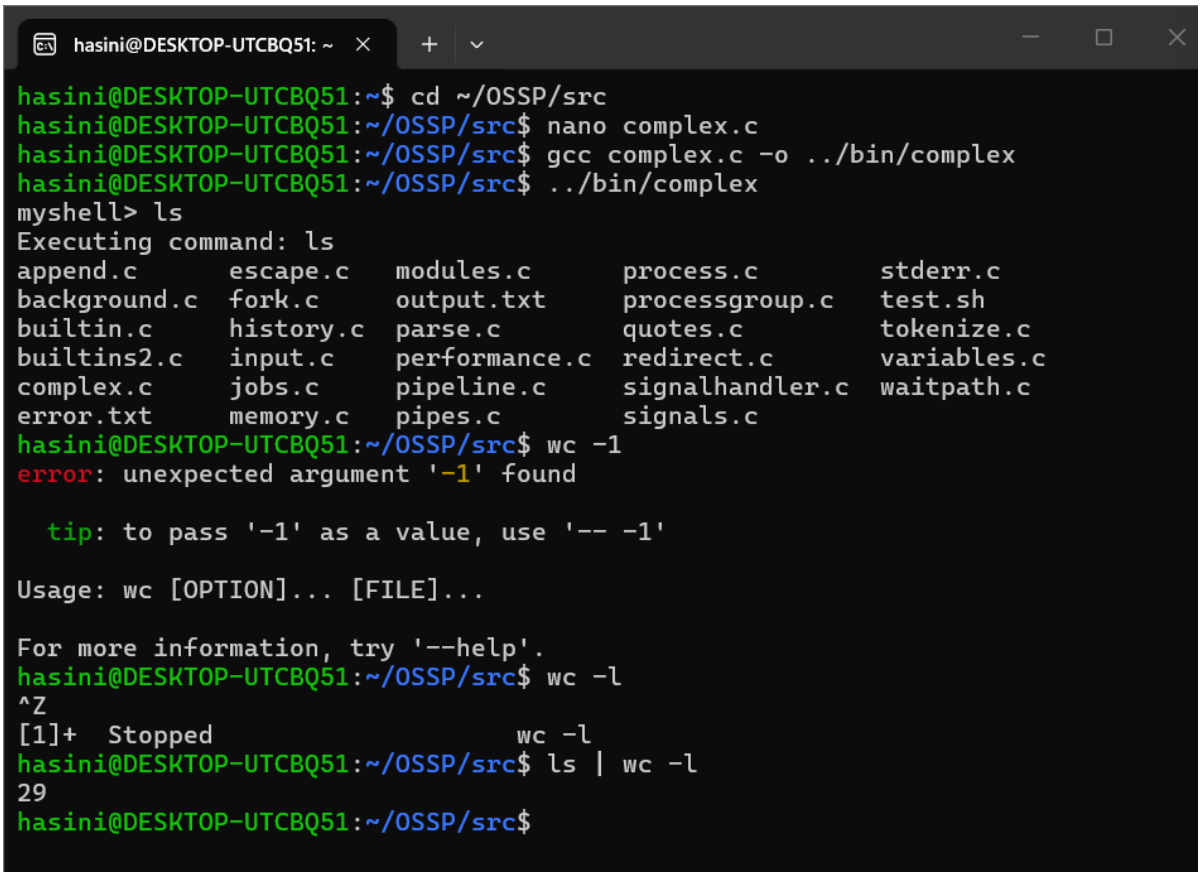
        if (pid1 == 0)
        {
            dup2(pipefd[1], STDOUT_FILENO);
            close(pipefd[0]);
            close(pipefd[1]);

            char *args[10];
            int i = 0;
            char *token = strtok(command1, " ");

            while (token != NULL)
            {
```

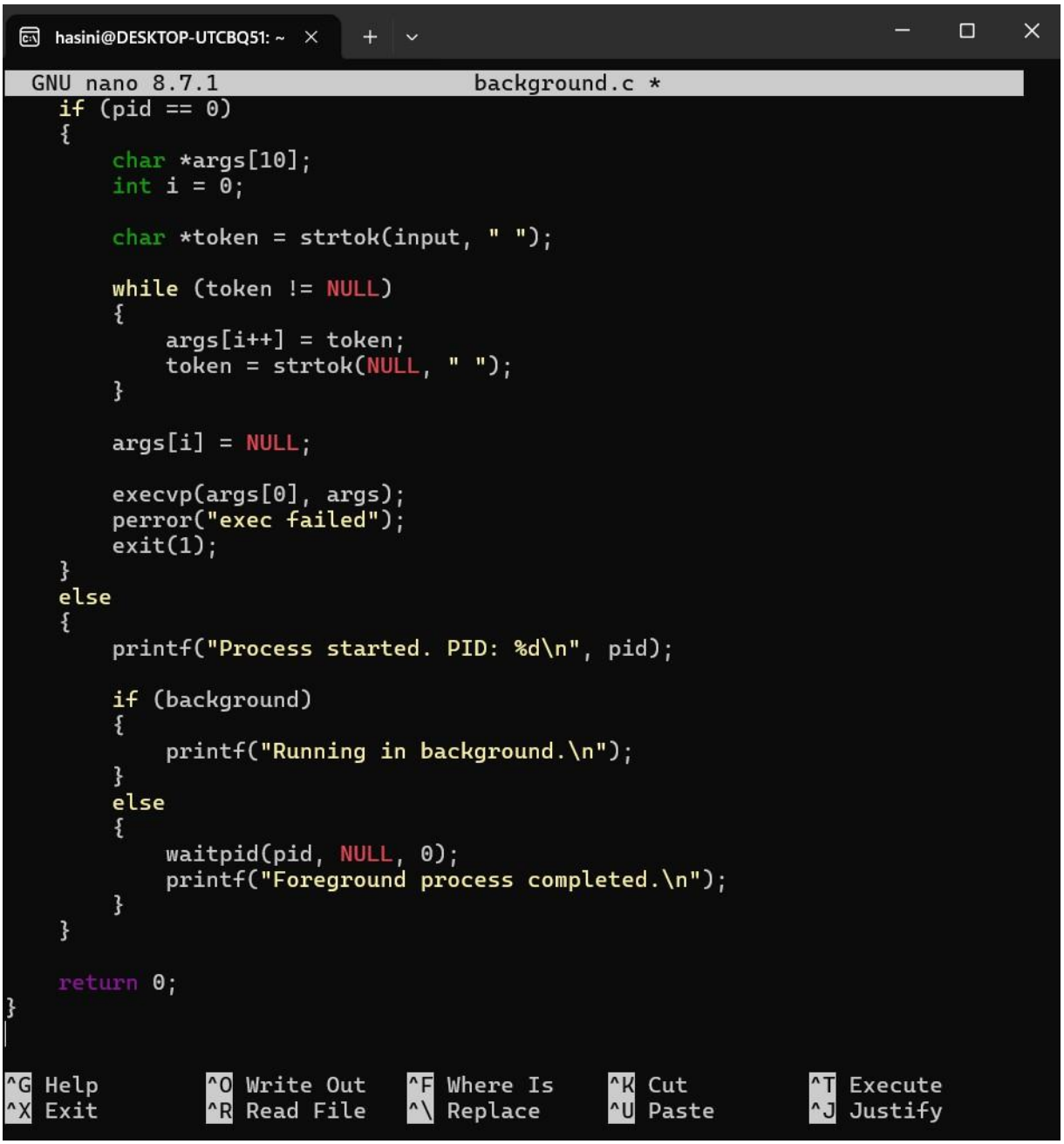

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 complex.c *
while (token != NULL)
{
args[i++] = token;
token = strtok(NULL, " ");
}
args[i] = NULL;
execvp(args[0], args);
perror("exec failed");
exit(1);
}
pid_t pid2 = fork();
if (pid2 == 0)
{
dup2(pipefd[0], STDIN_FILENO);
close(pipefd[1]);
close(pipefd[0]);
char *args[10];
int i = 0;
char *token = strtok(command2, " ");
while (token != NULL)
{
args[i++] = token;
token = strtok(NULL, " ");
}
args[i] = NULL;
execvp(args[0], args);
perror("exec failed");
exit(1);
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

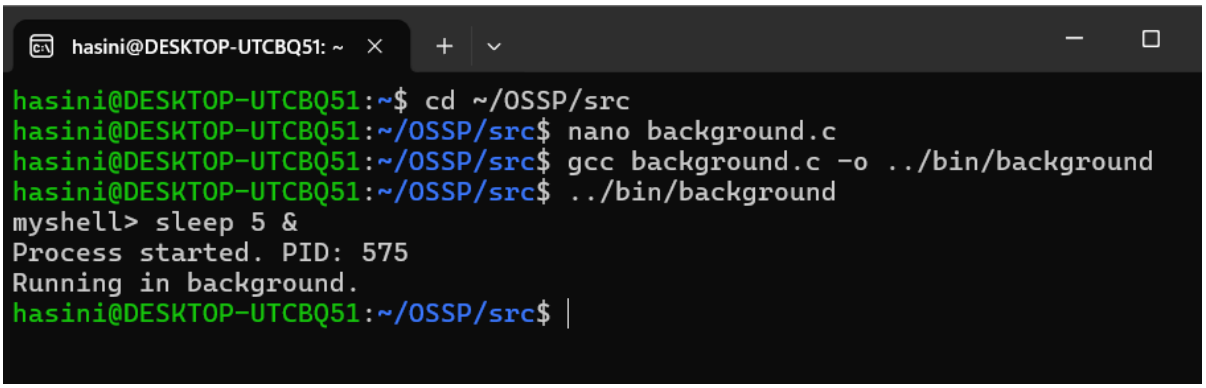

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 complex.c *
args[i] = NULL;
execvp(args[0], args);
perror("exec failed");
exit(1);
}
pid_t pid2 = fork();
if (pid2 == 0)
{
dup2(pipefd[0], STDIN_FILENO);
close(pipefd[1]);
close(pipefd[0]);
char *args[10];
int i = 0;
char *token = strtok(command2, " ");
while (token != NULL)
{
args[i++] = token;
token = strtok(NULL, " ");
}
args[i] = NULL;
execvp(args[0], args);
perror("exec failed");
exit(1);
}
close(pipefd[0]);
close(pipefd[1]);
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano complex.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc complex.c -o ../bin/complex
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/complex
myshell> ls
Executing command: ls
append.c escape.c modules.c process.c stderr.c
background.c fork.c output.txt processgroup.c test.sh
builtin.c history.c parse.c quotes.c tokenize.c
builtins2.c input.c performance.c redirect.c variables.c
complex.c jobs.c pipeline.c signalhandler.c waitpath.c
error.txt memory.c pipes.c signals.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ wc -1
error: unexpected argument '-1' found
tip: to pass '-1' as a value, use '-- -1'
Usage: wc [OPTION]... [FILE]...
For more information, try '--help'.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ wc -l
^Z
[1]+ Stopped wc -l
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ls | wc -l
29
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

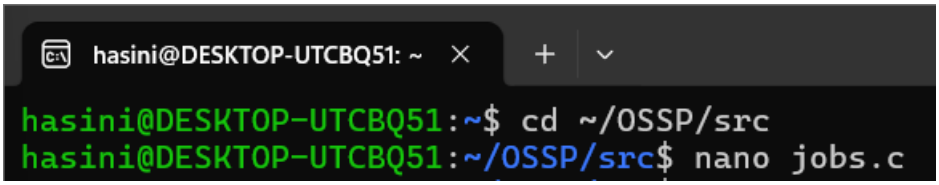
16.

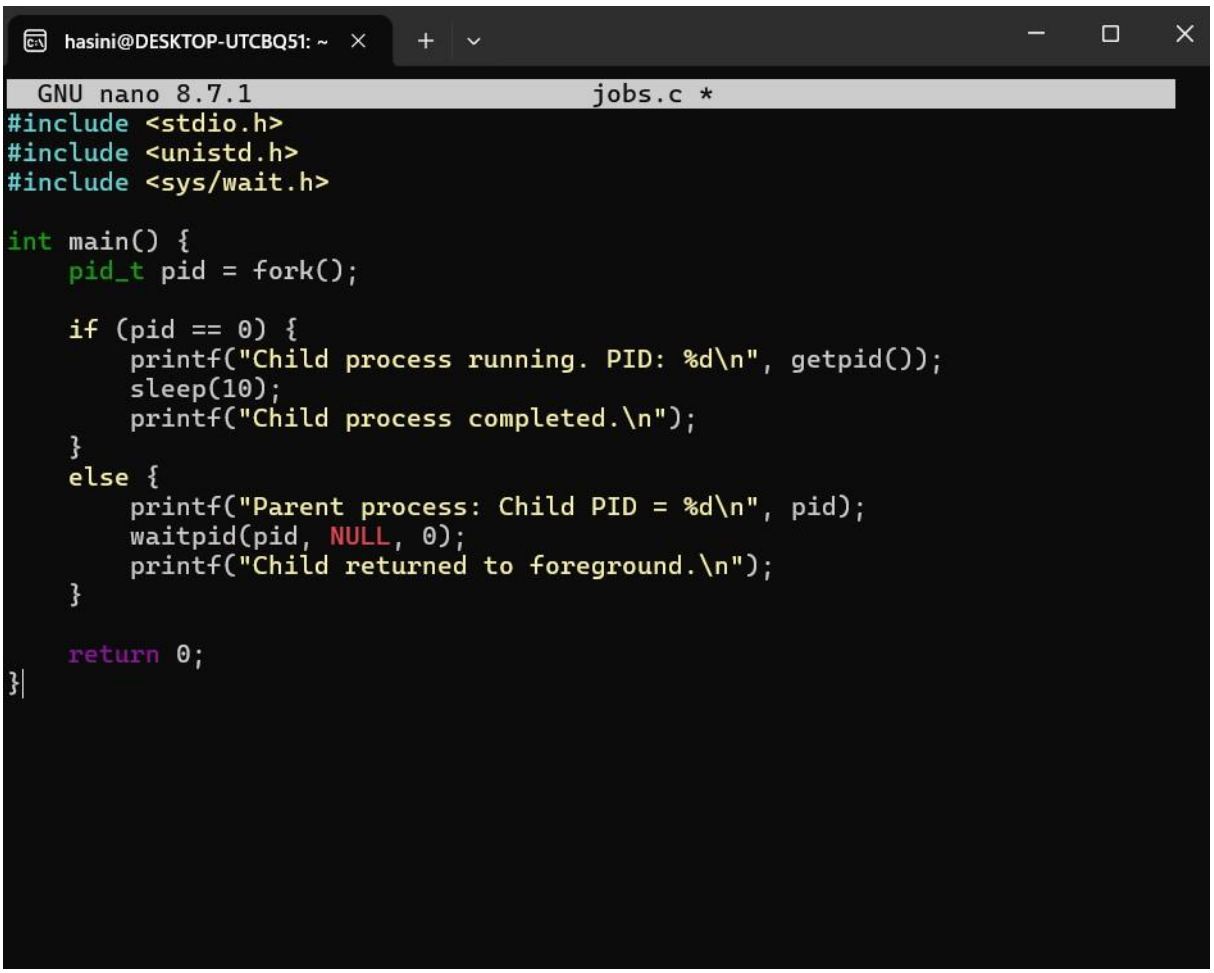

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 background.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
int main()
{
char input[100];
printf("myshell> ");
fgets(input, sizeof(input), stdin);
input[strcspn(input, "\n")] = '\0';
int background = 0;
int len = strlen(input);
if (len > 0 && input[len - 1] == '&')
{
background = 1;
input[len - 1] = '\0';
}
pid_t pid = fork();
if (pid == 0)
{
char *args[10];
int i = 0;
char *token = strtok(input, " ");
while (token != NULL)
{
args[i++] = token;
token = strtok(NULL, " ");
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

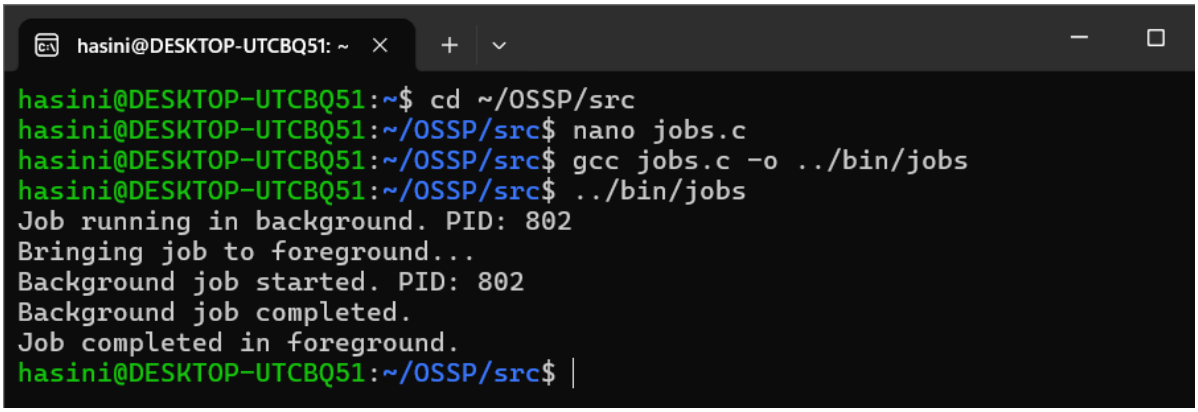

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 background.c *
if (pid == 0)
{
char *args[10];
int i = 0;
char *token = strtok(input, " ");
while (token != NULL)
{
args[i++] = token;
token = strtok(NULL, " ");
}
args[i] = NULL;
execvp(args[0], args);
perror("exec failed");
exit(1);
}
else
{
printf("Process started. PID: %d\n", pid);
if (background)
{
printf("Running in background.\n");
}
else
{
waitpid(pid, NULL, 0);
printf("Foreground process completed.\n");
}
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano background.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc background.c -o ../bin/background
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/background
myshell> sleep 5 &
Process started. PID: 575
Running in background.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

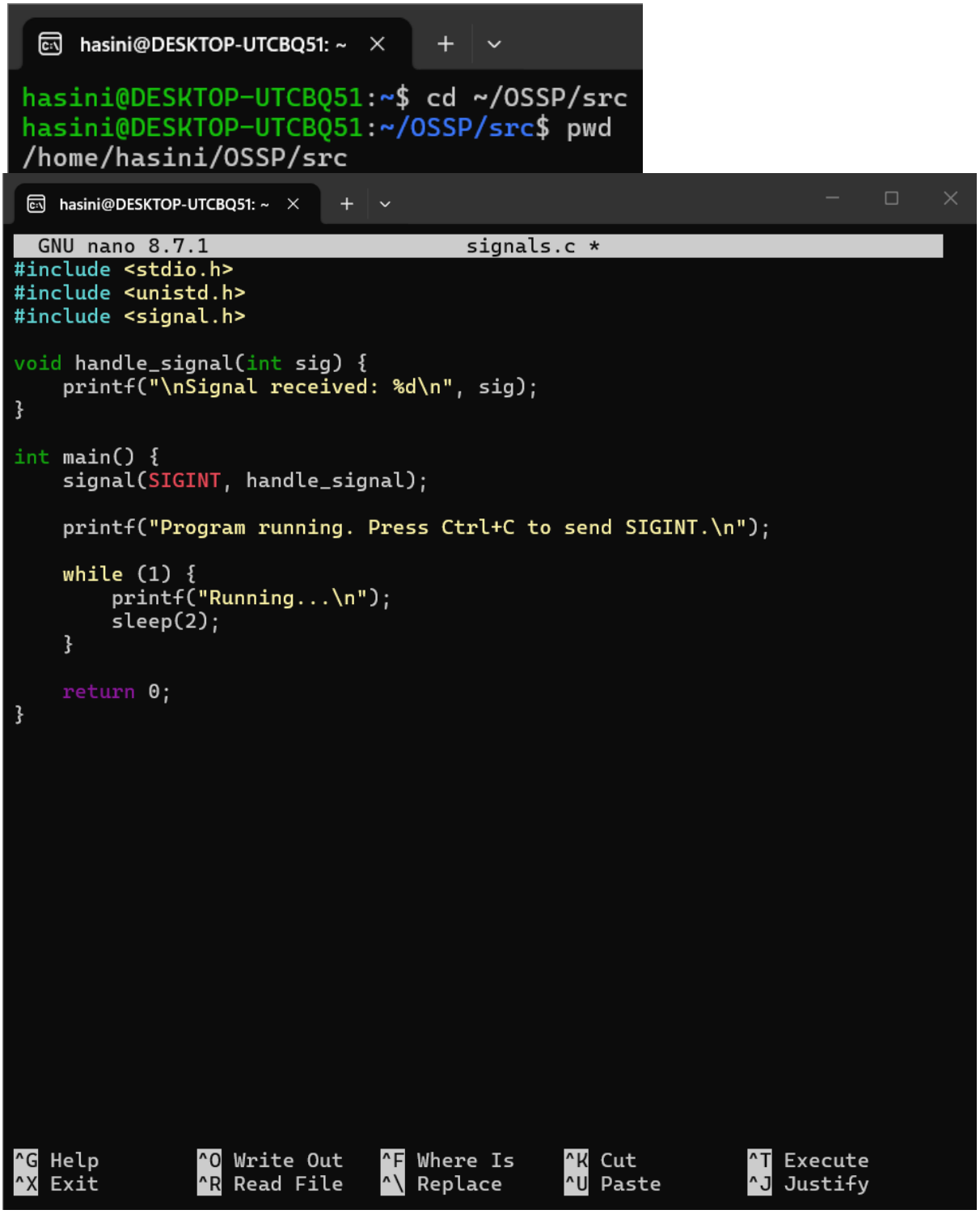
17.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano jobs.c

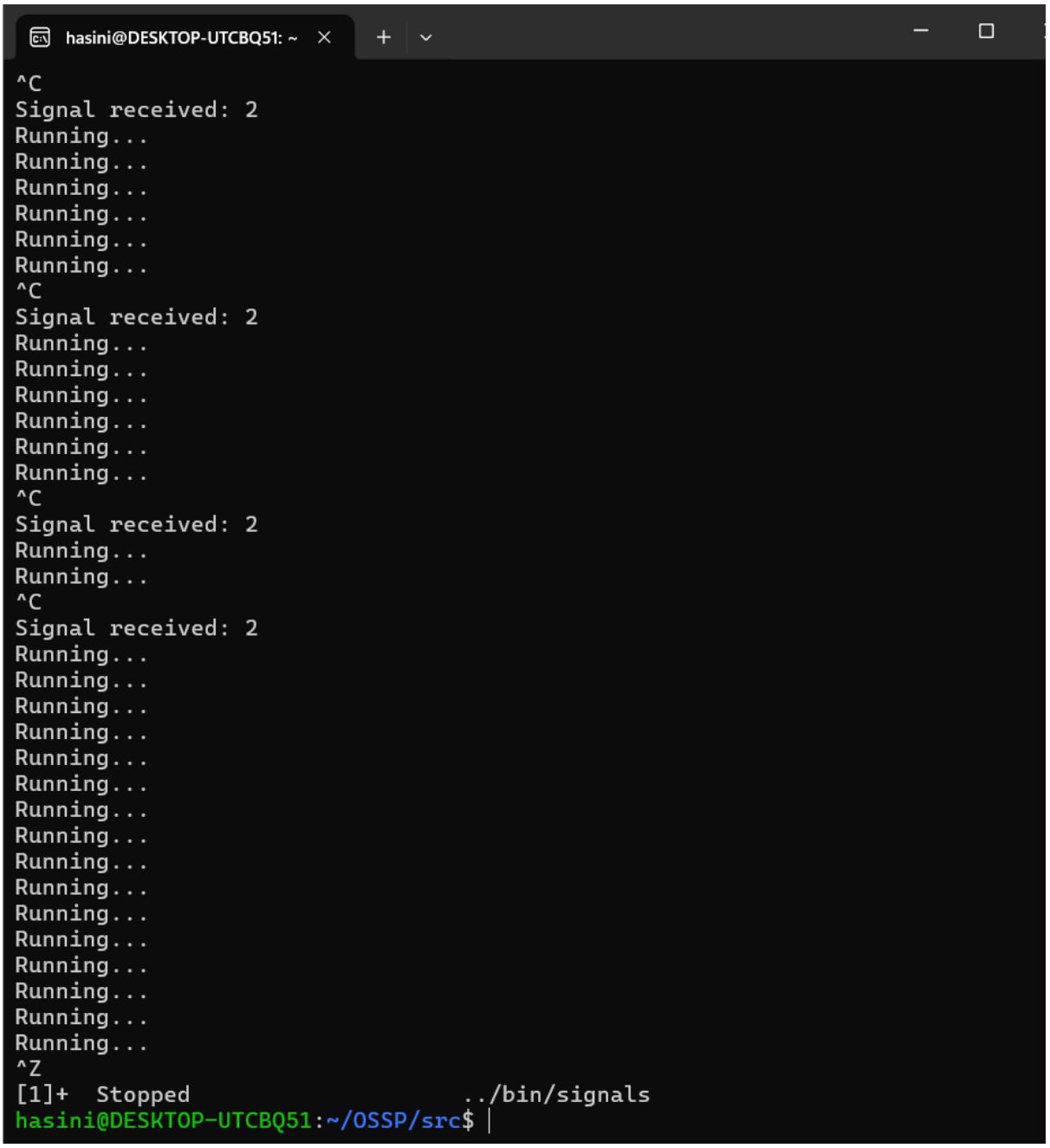

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 jobs.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
pid_t pid = fork();
if (pid == 0) {
printf("Child process running. PID: %d\n", getpid());
sleep(10);
printf("Child process completed.\n");
}
else {
printf("Parent process: Child PID = %d\n", pid);
waitpid(pid, NULL, 0);
printf("Child returned to foreground.\n");
}
return 0;
}


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano jobs.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc jobs.c -o ../bin/jobs
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/jobs
Job running in background. PID: 802
Bringing job to foreground...
Background job started. PID: 802
Background job completed.
Job completed in foreground.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

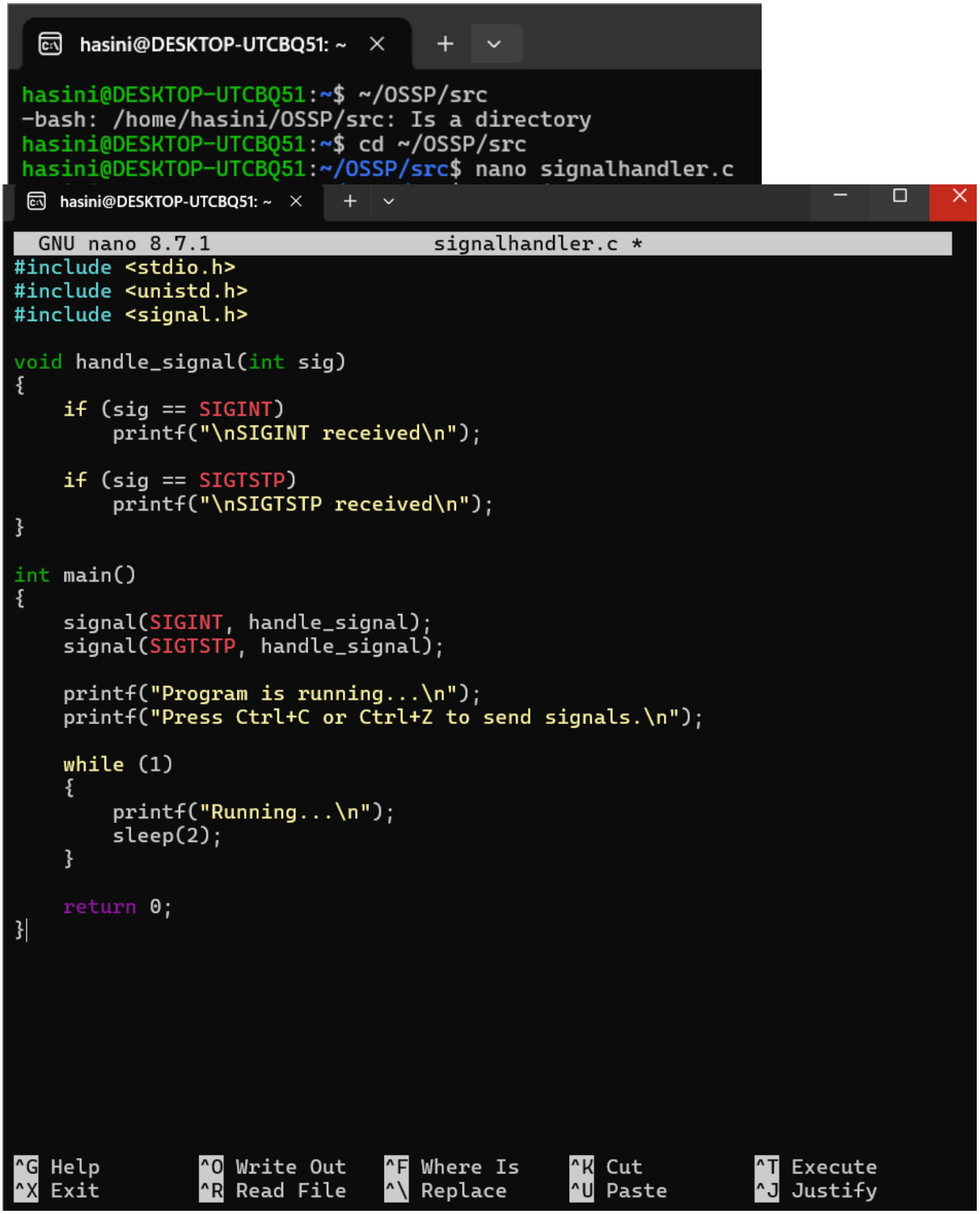
18.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ pwd
/home/hasini/OSSP/src
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 signals.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
void handle_signal(int sig) {
printf("\nSignal received: %d\n", sig);
}
int main() {
signal(SIGINT, handle_signal);
printf("Program running. Press Ctrl+C to send SIGINT.\n");
while (1) {
printf("Running...\n");
sleep(2);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

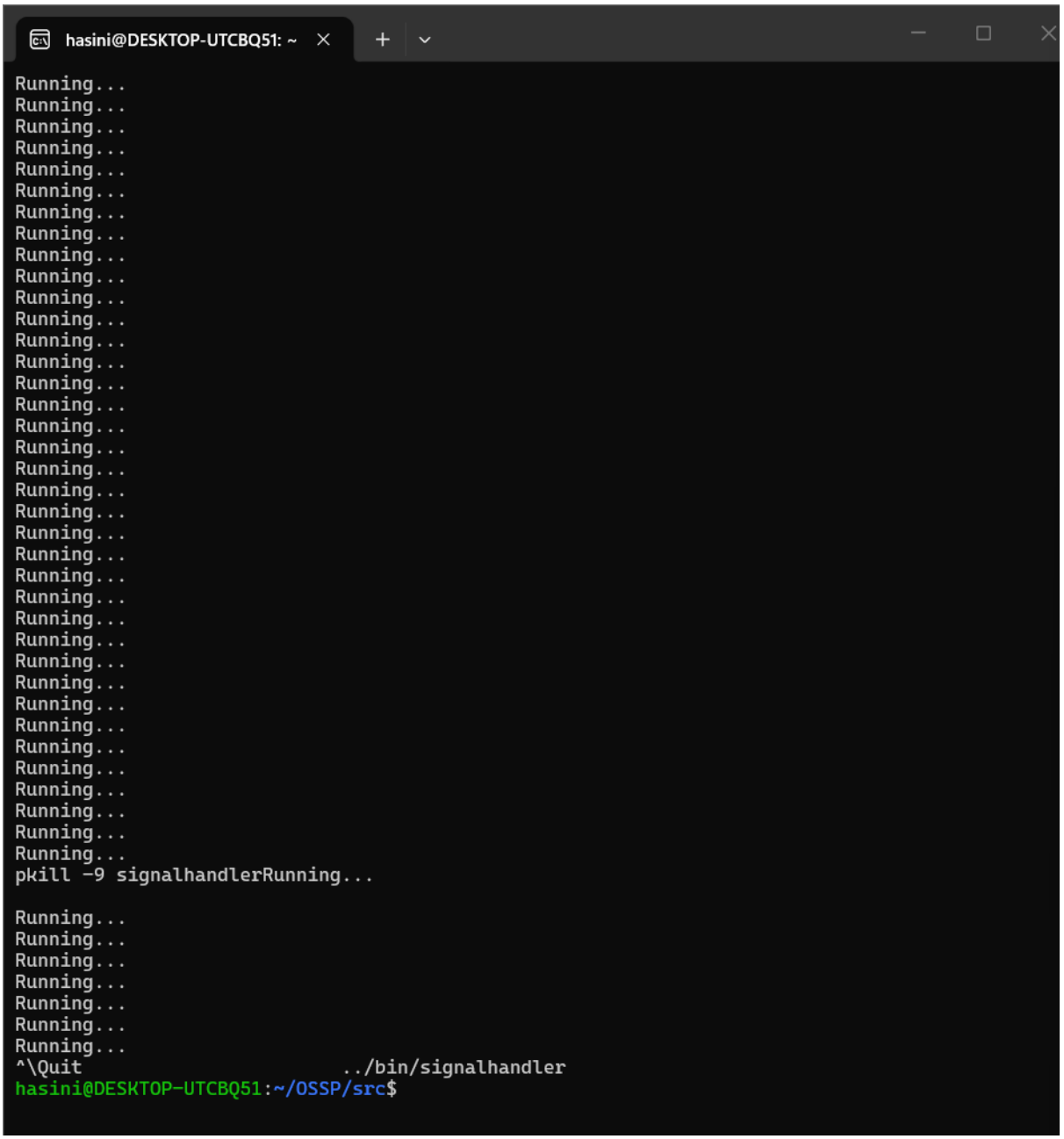
```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ mv ~/signals.c ~/OSSP/src/
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc signals.c -o ../bin/signals
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/signals
Program running. Press Ctrl+C to send SIGINT.
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
../bin/signalsRunning...
Running...
^C
Signal received: 2
Running...
Running...
Running...
^C
Signal received: 2
Running...
Running...
^C
Signal received: 2
Running...
Running...
```


hasini@DESKTOP-UTCBQ51: ~
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
^C
Signal received: 2
Running...
Running...
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
^Z
[1]+ Stopped ../bin/signals
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

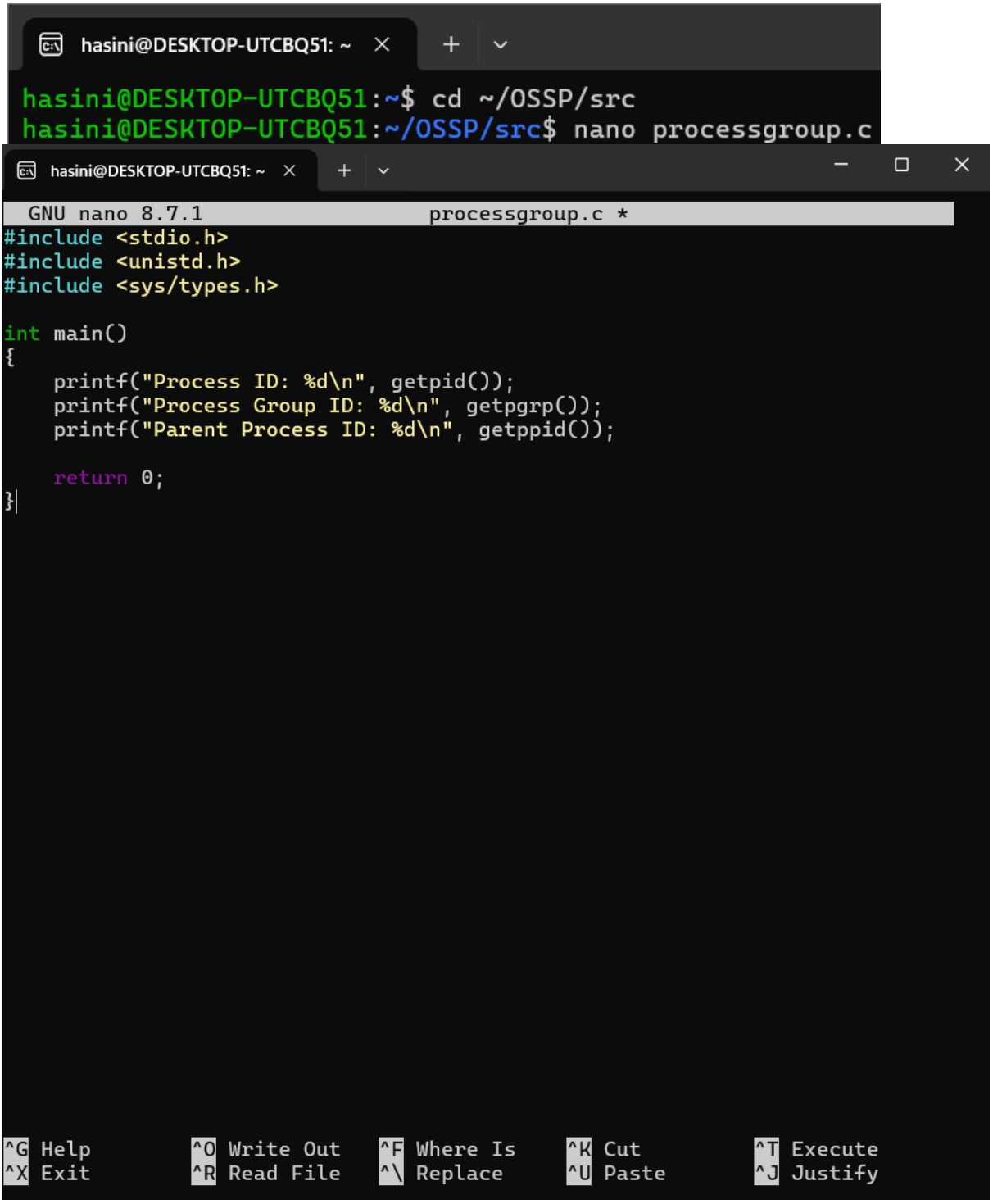
19.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano signalhandler.c
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 signalhandler.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
void handle_signal(int sig)
{
if (sig == SIGINT)
printf("\nSIGINT received\n");
if (sig == SIGTSTP)
printf("\nSIGTSTP received\n");
}
int main()
{
signal(SIGINT, handle_signal);
signal(SIGTSTP, handle_signal);
printf("Program is running...\n");
printf("Press Ctrl+C or Ctrl+Z to send signals.\n");
while (1)
{
printf("Running...\n");
sleep(2);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc signalhandler.c -o ../bin/signalhandler
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/signalhandler
Program is running...
Press Ctrl+C or Ctrl+Z to send signals.
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
Running...
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
^XRunning...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
killRunning...
 Running...
51
Running...
Running...
killRunning...
Running...
Running...
Running...
```

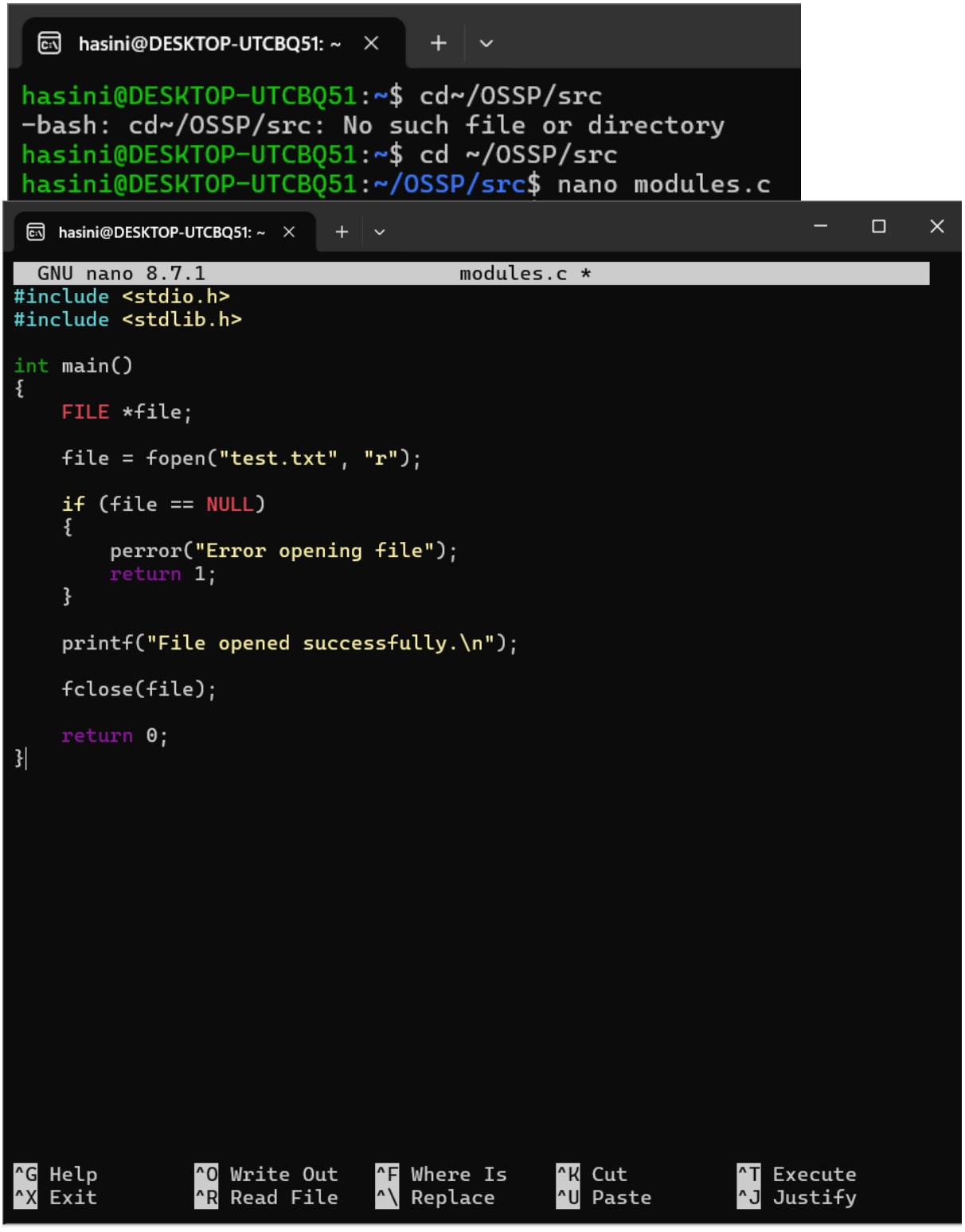

hasini@DESKTOP-UTCBQ51: ~
Running...
Running...
Running...

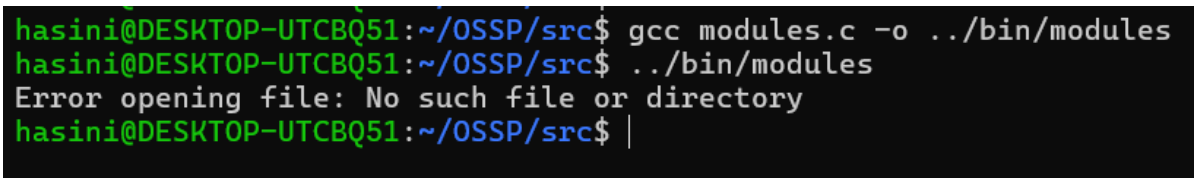
20.




```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc processgroup.c -o ../bin/processgroup
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/processgroup
Process ID: 791
Process Group ID: 791
Parent Process ID: 672
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

21.

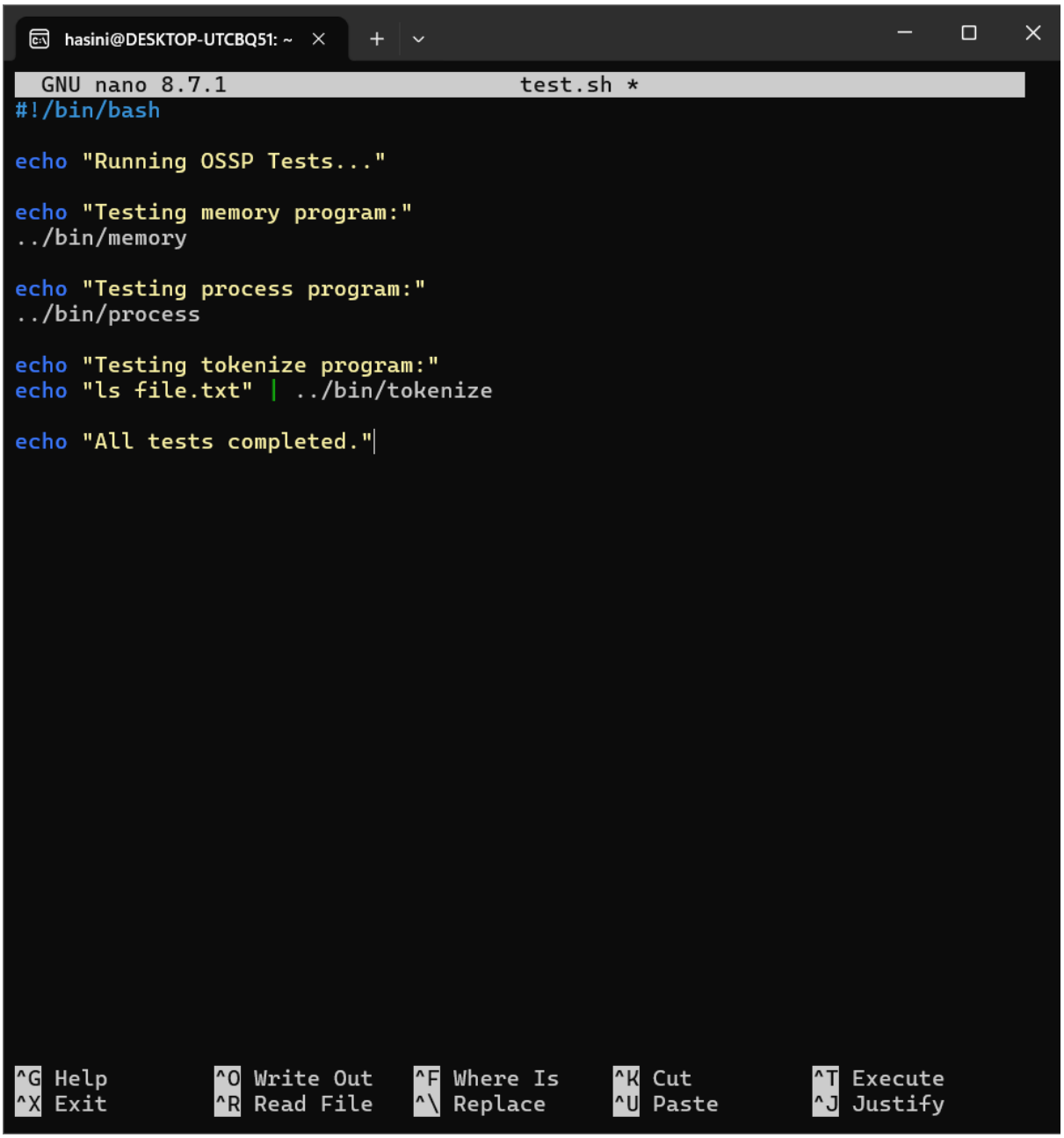

hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd~/OSSP/src
-bash: cd~/OSSP/src: No such file or directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano modules.c
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 modules.c *
#include <stdio.h>
#include <stdlib.h>
int main()
{
FILE *file;
file = fopen("test.txt", "r");
if (file == NULL)
{
perror("Error opening file");
return 1;
}
printf("File opened successfully.\n");
fclose(file);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc modules.c -o ../bin/modules
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/modules
Error opening file: No such file or directory
hasini@DESKTOP-UTCBQ51:~/OSSP/src$

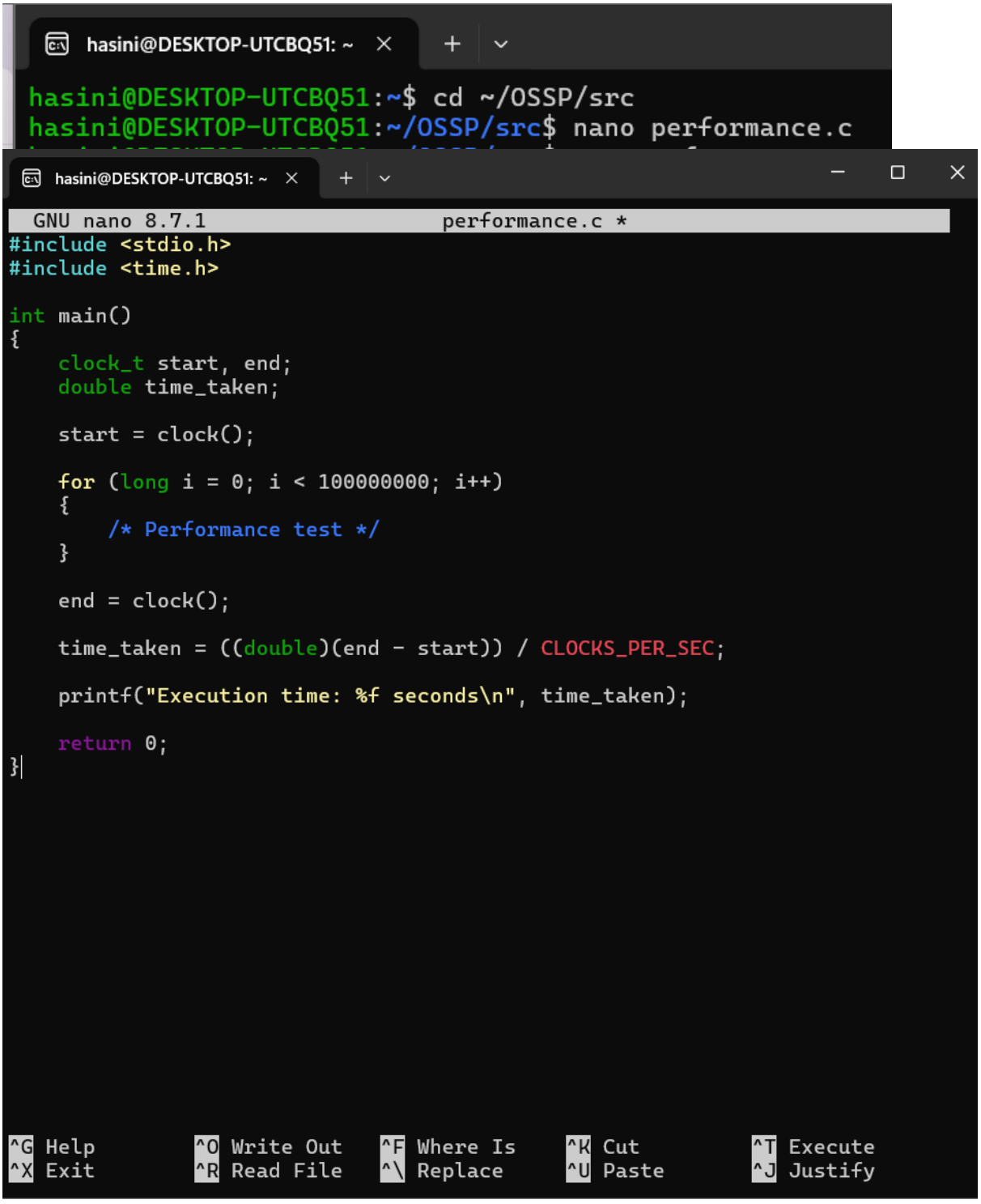
22. Valgrind and Test Scripts:

```
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
==827== Memcheck, a memory error detector
==827== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==827== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==827== Command: ../bin/memory
==827==
Data: Hello
Memory released successfully
==827==
==827== HEAP SUMMARY:
==827==     in use at exit: 0 bytes in 0 blocks
==827==   total heap usage: 2 allocs, 2 frees, 1,088 bytes allocated
==827==
==827== All heap blocks were freed -- no leaks are possible
==827==
==827== For lists of detected and suppressed errors, rerun with: -s
==827== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

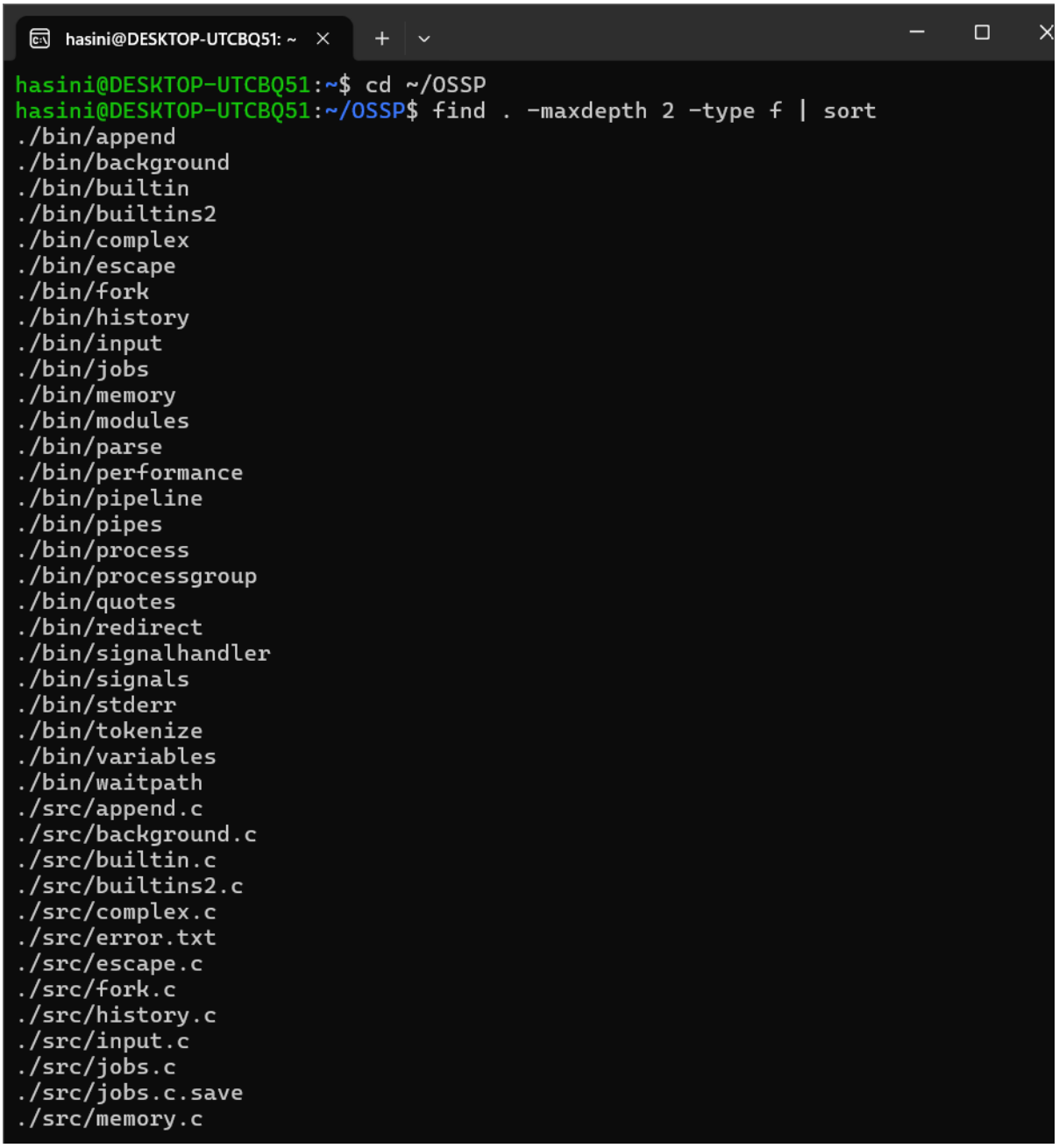
```
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano test.sh
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ chmod +x test.sh
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ./test.sh
Running OSSP Tests...
Testing memory program:
Data: Hello
Memory released successfully
Testing process program:
Process ID: 879
Parent Process ID: 877
Testing tokenize program:
Enter a command: Tokens:
ls
file.txt
All tests completed.
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

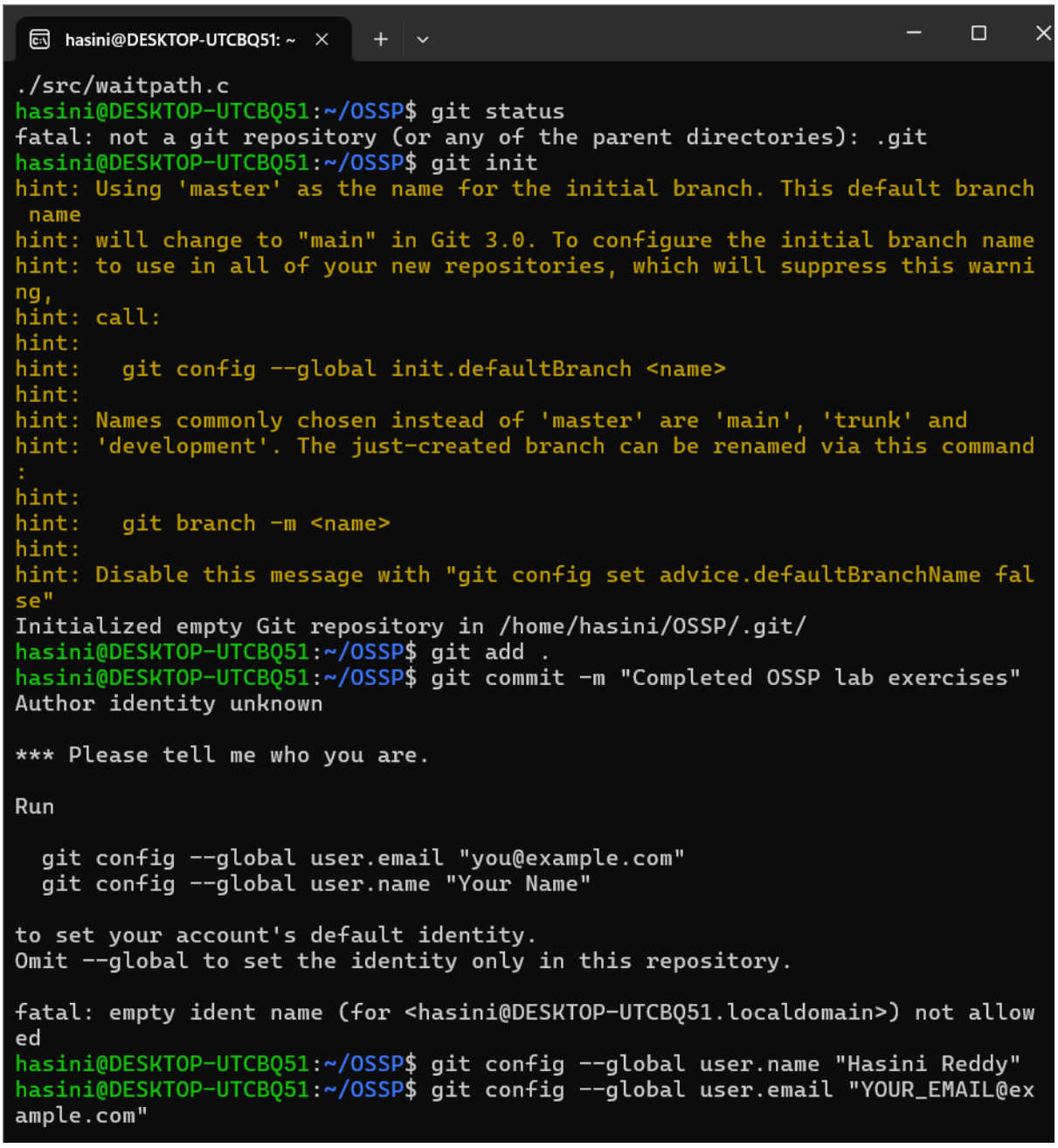

hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 test.sh *
#!/bin/bash
echo "Running OSSP Tests..."
echo "Testing memory program:"
../bin/memory
echo "Testing process program:"
../bin/process
echo "Testing tokenize program:"
echo "ls file.txt" | ../bin/tokenize
echo "All tests completed."
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

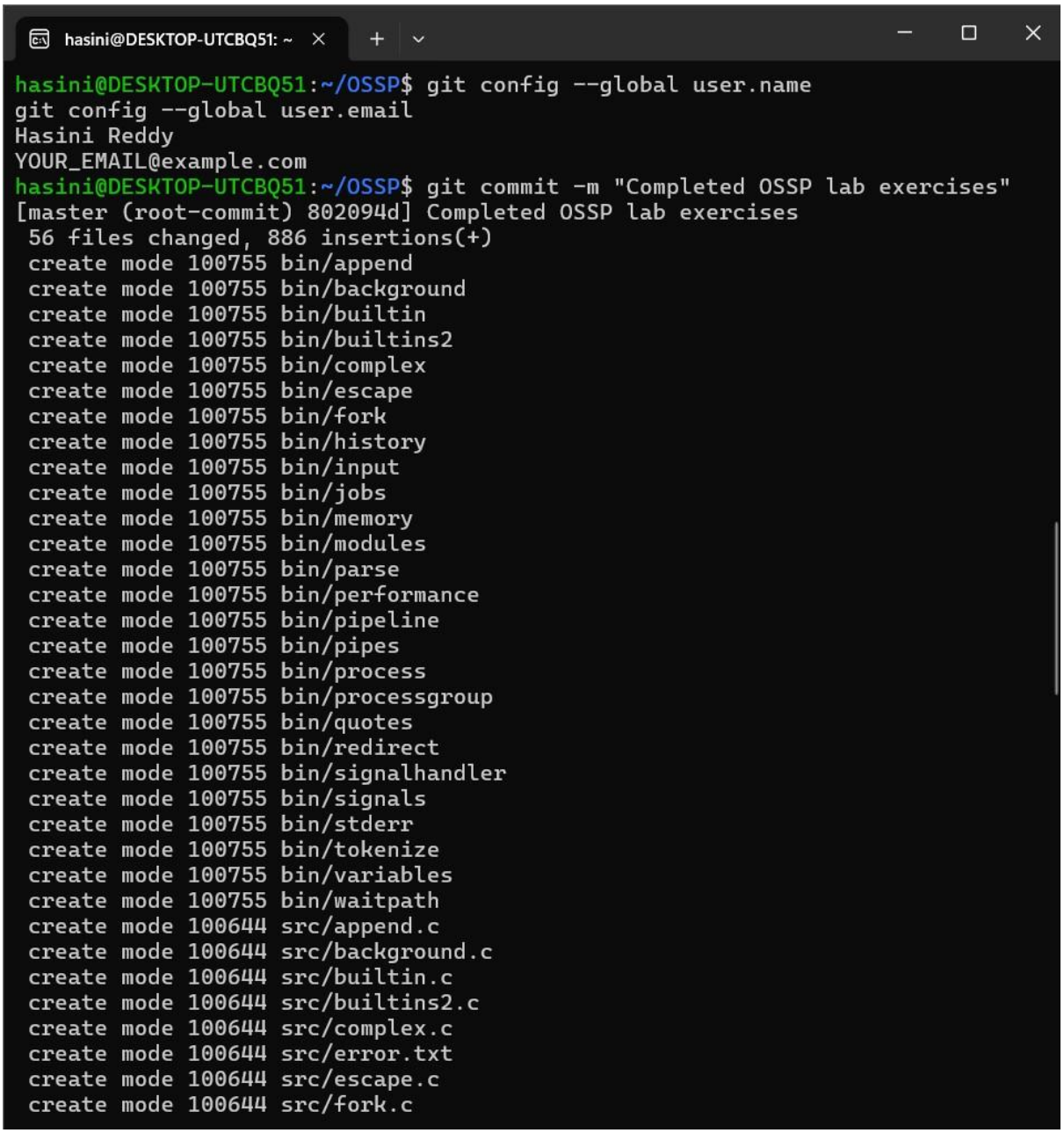
23.

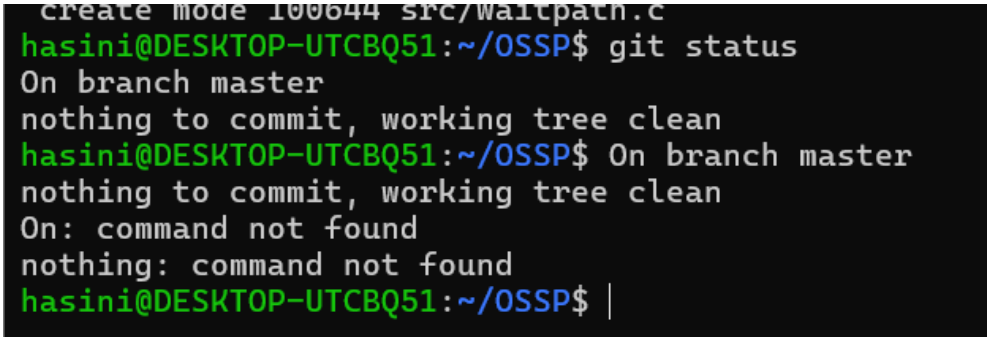

hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano performance.c
hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 performance.c *
#include <stdio.h>
#include <time.h>
int main()
{
clock_t start, end;
double time_taken;
start = clock();
for (long i = 0; i < 100000000; i++)
{
/* Performance test */
}
end = clock();
time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;
printf("Execution time: %f seconds\n", time_taken);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

24.


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP
hasini@DESKTOP-UTCBQ51:~/OSSP$ find . -maxdepth 2 -type f | sort
./bin/append
./bin/background
./bin/builtin
./bin/builtins2
./bin/complex
./bin/escape
./bin/fork
./bin/history
./bin/input
./bin/jobs
./bin/memory
./bin/modules
./bin/parse
./bin/performance
./bin/pipeline
./bin/pipes
./bin/process
./bin/processgroup
./bin/quotes
./bin/redirect
./bin/signalhandler
./bin/signals
./bin/stderr
./bin/tokenize
./bin/variables
./bin/waitpath
./src/append.c
./src/background.c
./src/builtin.c
./src/builtins2.c
./src/complex.c
./src/error.txt
./src/escape.c
./src/fork.c
./src/history.c
./src/input.c
./src/jobs.c
./src/jobs.c.save
./src/memory.c


hasini@DESKTOP-UTCBQ51: ~
./src/waitpath.c
hasini@DESKTOP-UTCBQ51:~/OSSP$ git status
fatal: not a git repository (or any of the parent directories): .git
hasini@DESKTOP-UTCBQ51:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/hasini/OSSP/.git/
hasini@DESKTOP-UTCBQ51:~/OSSP$ git add .
hasini@DESKTOP-UTCBQ51:~/OSSP$ git commit -m "Completed OSSP lab exercises"
Author identity unknown
*** Please tell me who you are.
Run
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
to set your account's default identity.
Omit --global to set the identity only in this repository.
fatal: empty ident name (for <hasini@DESKTOP-UTCBQ51.localdomain>) not allow
ed
hasini@DESKTOP-UTCBQ51:~/OSSP$ git config --global user.name "Hasini Reddy"
hasini@DESKTOP-UTCBQ51:~/OSSP$ git config --global user.email "YOUR_EMAIL@ex
ample.com"


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~/OSSP$ git config --global user.name
git config --global user.email
Hasini Reddy
YOUR_EMAIL@example.com
hasini@DESKTOP-UTCBQ51:~/OSSP$ git commit -m "Completed OSSP lab exercises"
[master (root-commit) 802094d] Completed OSSP lab exercises
56 files changed, 886 insertions(+)
create mode 100755 bin/append
create mode 100755 bin/background
create mode 100755 bin/builtin
create mode 100755 bin/builtins2
create mode 100755 bin/complex
create mode 100755 bin/escape
create mode 100755 bin/fork
create mode 100755 bin/history
create mode 100755 bin/input
create mode 100755 bin/jobs
create mode 100755 bin/memory
create mode 100755 bin/modules
create mode 100755 bin/parse
create mode 100755 bin/performance
create mode 100755 bin/pipeline
create mode 100755 bin/pipes
create mode 100755 bin/process
create mode 100755 bin/processgroup
create mode 100755 bin/quotes
create mode 100755 bin/redirect
create mode 100755 bin/signalhandler
create mode 100755 bin/signals
create mode 100755 bin/stderr
create mode 100755 bin/tokenize
create mode 100755 bin/variables
create mode 100755 bin/waitpath
create mode 100644 src/append.c
create mode 100644 src/background.c
create mode 100644 src/builtin.c
create mode 100644 src/builtins2.c
create mode 100644 src/complex.c
create mode 100644 src/error.txt
create mode 100644 src/escape.c
create mode 100644 src/fork.c


create mode 100644 src/waitpath.c
hasini@DESKTOP-UTCBQ51:~/OSSP$ git status
On branch master
nothing to commit, working tree clean
hasini@DESKTOP-UTCBQ51:~/OSSP$ On branch master
nothing to commit, working tree clean
On: command not found
nothing: command not found
hasini@DESKTOP-UTCBQ51:~/OSSP$